

Mesoscale FEM Model of Concrete: Statistical Assessment of Inherent Stress Concentrations in Dependence on Phase Heterogeneity

Jan Mašek^{a,b}, Petr Miarka^{a,b}

^a*Institute of Physics of Materials,
Czech Academy of Sciences, Žitkova 22, 616 00 Brno, Czech Republic,
e-mail: masek@ipm.cz, miarka@ipm.cz*

^b*Institute of Structural Mechanics,
Faculty of Civil Engineering, Brno University of Technology, Veveří 331/95, 602 00 Brno, Czech Republic,
e-mail: jan.masek1@vut.cz, petr.miarka@vut.cz*

Abstract

Concrete heterogeneity originates from its production process, which involves bonding aggregates with a binder matrix. This study presents a mesoscale finite element model (MFEM) that offers detailed insights into the fracture process at the aggregate–cement matrix interface, focusing on one of concrete’s key properties: its mechanical response. Unlike discrete models, which often average out critical stress concentrations within the mesostructure, the MFEM approach captures detailed stress distributions, revealing localized effects crucial for understanding damage evolution.

Although computationally more demanding, the MFEM leverages modern high-performance computing (HPC) to provide a detailed description of the stress field and material damage across different phases and interfaces.

The proposed modeling framework integrates a collision-checked aggregate generation procedure, Voronoi-based mesostructure construction, and adaptive 3D meshing, forming a reusable methodology for stress analysis in heterogeneous composites. This approach offers transparent, physically interpretable parameterization of phase properties in contrast to black-box discrete models.

Another methodological contribution is the statistical post-processing of stress data using histogram-based analysis across cross-sectional planes. This enables quantitative evaluation of stress concentration distributions, providing valuable insights into the mesoscale mechanical response and serving as a useful visualization tool for researchers working on heterogeneous material modeling.

Various matrix-to-aggregate stiffness ratios are considered to evaluate the influence of material heterogeneity on the stress field. The results are based on a statistical evaluation of stress concentrations arising from variations in material stiffness. The model is applied to investigate the impact of using recycled crushed bricks as aggregates in concrete, with particular emphasis on the stiffness mismatch between the matrix and aggregates. The study examines how this stiffness contrast affects stress distribution and ultimately influences the composite’s failure mechanisms. Beyond this application, the MFEM framework provides a foundation for further investigations into nonlinear fracture processes, fatigue analysis, and mechanical optimization of alternative aggregate-matrix systems.

Keywords:

MFEM, meso-scale, concrete, heterogeneity, aggregate-cement interface, stiffness mismatch, recycled aggregates

Publishing information

Mašek J., Miarka P., *FEM model of concrete: Statistical assessment of inherent stress concentrations in dependence on phase heterogeneity*, Finite Elements in Analysis and Design, 2025, 252, 104442, ISSN 0168-874X
DOI: <https://doi.org/10.1016/j.finel.2025.104442>.

Copyright: ©2024. This version is made available under the CC-BY 4.0 license and all right remains to authors according to CZ/EU legislation. <https://creativecommons.org/licenses/by/4.0/>

1. Introduction

Concrete is one of the most widely used construction materials worldwide due to its structural versatility, shape variability, and long-term durability. From a continuum mechanics perspective, concrete is a heterogeneous material, with the origin of its heterogeneity rooted in the production process, which involves binding aggregates together with a binder—typically granite aggregates combined with Portland cement [1]. In structural applications, however, concrete is usually treated as a homogeneous material, with its mechanical properties defined according to established standards [2].

Nevertheless, this inherent heterogeneity becomes apparent at smaller scales, where it can significantly influence the initiation of damage. More specifically, concrete is a multi-phase material, with the number and nature of its phases depending on the scale under investigation [3]. At the aggregate scale, three distinct phases can be identified: aggregates, cement paste, and air pores. Additionally, a fourth phase—the interfacial transition zone (ITZ)—represents the bonding behavior between the aggregates and the cement paste [4]. The ITZ plays a significant role in damage initiation due to high porosity and bonding capability below the plain cement matrix. ITZ is indeed an intricate region to examine mechanically, due to its small scale, heterogeneity, and complex microstructure, see the review in [5]. The differences in concrete material scales are illustrated in Figure 1.

The number of phases depends on the investigated scale. For structural applications, it is beneficial to use homogenized behavior, whereas for processes occurring at the aggregate–cement matrix level, a multi-phase material model is necessary to accurately describe the observed behavior [6]. This inherent heterogeneity poses a challenge for accurately calculating stresses, which are crucial for predicting future failure under both static and cyclic loading conditions.

Significant effort has been invested in developing numerical models of the mechanical response of concrete and formulating constitutive laws, which allow for more accurate predictions of stress and subsequent damage. The most common material models are typically formulated in tensorial form within the classical continuum theory, such as the Concrete Damage Plasticity Model [7], the Fracture-Plastic Constitutive Model [8] implemented in ATENA software, microplane models [9, 10], and phase-field models [11]. These models treat the material as macroscopically homogeneous without explicitly accounting for lower-scale heterogeneity.

In contrast to the constitutive material models, aggregate-resolution models (meso-scale models) models of concrete directly involve individual heterogeneous units [12], which arise from the combination of a constitutive relation at a mesoscale and explicitly modeled material structure [13]. A meso-scale model can provide information about the fracture processes at the level of the aggregate-to-cement matrix bond [14]. Thus, it can properly describe the most prominent concrete disadvantage, i.e., the strain-softening behavior as a response to tensile load [4]. Besides continuum-based models, there exist multiple other solutions such as the lattice discrete particle model (LDPM) [15] or discrete finite element method (DFEM) [16], where the inter-connectivity of each particle has its defined interaction. Phase-field models represent fractures as diffuse damage zones, simplifying crack initiation and propagation without explicit tracking. They are widely used for simulating complex fracture patterns in heterogeneous materials at the aggregate scale. However, these models can be computationally intensive and may have limitations in accurately capturing sharp crack interfaces and fine-scale material heterogeneity [17].

The discrete models, particularly LDPM, often rely on empirical contact laws and parameter calibration, limiting their interpretability and generalizability. In contrast, aggregate-resolving finite element method (MFEM) models explicitly resolve the internal geometry and mechanical properties of each phase, providing physically interpretable fields of stress and strain and allowing direct control over material heterogeneity and microstructure geometry.

On the other hand, the use of aggregate-resolution MFEM models allows capturing the internal physical mech-

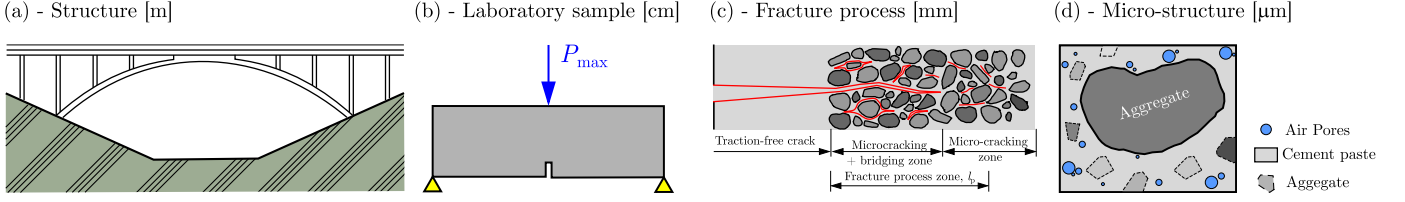


Figure 1: Scales of analysis of concrete structures.

anisms of cracking as well as the macro-scale structural response. We refer the reader to models developed in 3D in [18] and 2D in [19], respectively. 3D mesoscale models can be used for simulating reinforced concrete's resistance as well as ballistic penetration [20]. Also, in conjunction with cohesive elements, aggregate-resolving MFEM models can simulate the development of microcracking within the fracture process zone, see e.g. [21]. The present work accompanies these concepts by developing a fully 3D, collision-checked aggregate placement and meso-geometry framework combined with adaptive meshing, constituting a reusable methodology for simulating stress distributions and damage evolution in heterogeneous composites beyond concrete alone. Although MFEM is computationally more intensive than discrete models or continuum homogenized models, leveraging high-performance computing (HPC) and automation tools makes real-scale simulations feasible.

This combination of geometric fidelity, physical transparency, and computational practicality positions MFEM as a versatile, reusable modeling framework applicable not only to concrete but also to other heterogeneous materials where mesoscale structure strongly influences mechanical response.

The engineering practice often desires simpler solutions than using sophisticated and often computationally expensive models. The structural design of concrete load-bearing structures typically uses well-acknowledged linear elastic beam theory, often called Euler-Bernoulli's theory, for stress and deformation calculations, unfortunately without accounting for the high heterogeneity of concrete. The stress is then adjusted based on design recommendations in standards founded on empirical or experimental observations [22]. Figure 2 compares bending stress calculated using various methods. It shows an example of how different computational approaches can simplify the stress calculation.

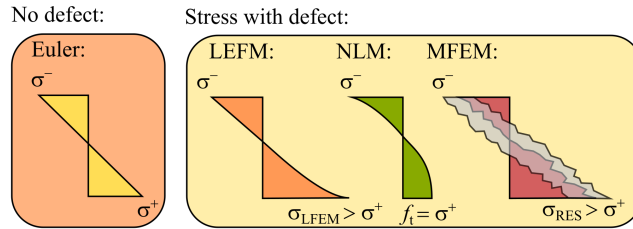


Figure 2: Illustration of normal stress caused by bending obtained by multiple calculation methods.

As shown in Figure 2, once the analyzed component contains a defect, the stress calculation must be adjusted to account for stress distribution changes along the height. Stress concentration arising from this defect can be addressed using the linear elastic fracture mechanics (LEFM) approach [23], which is commonly applied in studies of metals due to strong experimental support [24] and extensive documentation in handbooks.

In the case of nonlinear material (NLM) behavior, the resulting stress at the defect location does not exceed the material's tensile strength [7], and the damage is distributed (smeared) over the region of high stresses [25]. After reaching the tensile strength f_t , smeared cracking behavior is assumed, characterized by a maximum tensile strain ε_t usually related to the crack width w_c . In contrast, the MFEM model accounts for local stress concentrations not

only from the defect but also from the aggregate's shape and mechanical properties.

This numerical study investigates mechanical stress distribution through 3D MFEM simulations under different loading conditions and model geometries. These include three-point bending tests on both notched and unnotched beams to induce bending stresses, as well as a compressed cylinder representing a pure compression state.

Various matrix-to-aggregate stiffness ratios are considered to evaluate the influence of material heterogeneity on the stress field. The results are based on a statistical evaluation of stress concentrations arising from variations in material stiffness.

2. Motivation

The use of crushed brick as a replacement for natural aggregates in concrete offers both environmental and economic advantages. However, concerns remain regarding its mechanical performance, particularly due to the stiffness mismatch between brick aggregates and the cement matrix. Unlike granite aggregate concrete—where a uniform stiffness is often assumed based on empirical reliability—brick aggregate concrete exhibits a pronounced contrast in elastic properties. This discrepancy leads to stress concentrations at the aggregate-matrix interfaces, which can accelerate microcrack initiation and alter the overall stress distribution within the material [26, 27]. Understanding these variations in the stress field is crucial, as they directly influence crack propagation mechanisms and ultimately affect the structural integrity of the composite.

Recycled materials, including crushed bricks, are increasingly employed as coarse aggregates in concrete [28]. The incorporation of aggregates with mechanical properties significantly different from traditional materials such as granite can substantially impact the mechanical response of the composite [29]. In particular, stress concentration zones become critical, serving as initiation points for material damage and failure. Brick aggregates present additional complexities due to their inherent variability. Their material properties can vary considerably [30], with reported elastic modulus values ranging from 3.5 GPa to 34 GPa and Poisson's ratios between 0.12 and 0.29. Moreover, the crushing process may introduce further degradation, increasing heterogeneity among individual aggregates. This variability complicates the mechanical behavior of brick aggregate concrete, necessitating a deeper investigation into its stress distribution and failure mechanisms.

The purpose of the numerical study presented is to deliver detailed information about the internal stress distribution within concrete containing non-traditional aggregate materials. While the general effects of stiffness mismatch may have been predicted earlier, the scale of this study and the extensive statistical analysis of stress data provide a novel and unprecedented level of insight. In long-term, high-cycle fatigue, fracture originates from highly singular stress concentrations within a material volume otherwise well within its elastic regime. Therefore, the crack propagation and damage evolution is not the aim of the study - the attention is devoted to the influence of using various aggregate materials on inducing stress concentrations that will eventually trigger a damage.

3. Modeling approach

Several approaches have been developed for creating and placing the aggregates within the mesostructure, notably e.g. using the Voronoi tessellation [31, 32] or the Take-and-Place procedure [33]. The approach used in the present work is based on a weighted Voronoi tessellation and its adjustments for meso-structure generation informed by the material's sieve curve. Next, the numerical model is presented, including geometry, topology, and FE mesh generation.

3.1. Meso-structure topology generation

To model the heterogeneous structure of classical concrete, we first focus on capturing the shape and distribution of the aggregates. Figure 3a shows a typical concrete mesostructure containing granite aggregates embedded in a cement matrix. The shape of the aggregates depends on the material and origin of the aggregate source. However, aggregates often resemble sharp-edged convex polyhedra (e.g., crushed granite or crushed brick recycle). Therefore, without excessive simplification, it is reasonable to utilize convex polyhedral shapes within numerical models. Figure 3b illustrates how convex polyhedra can be identified in the depicted concrete mesostructure.



Figure 3: a) A typical concrete mesostructure, b) a possible approximation of aggregates by convex polyhedra.

The generation of the mesostructure begins with conducting a tessellation on a set of generating nodes. All figures in Figure 4 present a 2D illustration of the mesostructure generation process. A classic Voronoi cell, C_i , contains all domain points, $\mathbf{u}$, that are closer to the respective generating node $\mathbf{X}_i$ than to any other generating node $\mathbf{X}_k$ in the model domain, Ω :

$$C_i = \{\mathbf{u} \in \Omega \mid \forall k \neq i \quad \|\mathbf{u} - \mathbf{X}_i\| < \|\mathbf{u} - \mathbf{X}_k\|\} \quad (1)$$

Unlike the classic Voronoi tessellation, we utilize the Power diagram variant [34]. The Power diagram is a weighted version of the standard Voronoi tessellation, where each cell, P_i , encompasses all domain points associated with a sphere of radius r_i around its generating node $\mathbf{X}_i$:

$$P_i = \{\mathbf{u} \in \Omega \mid \forall k \neq i \quad \|\mathbf{u} - \mathbf{X}_i\|^2 - r_i^2 < \|\mathbf{u} - \mathbf{X}_k\|^2 - r_k^2\} \quad (2)$$

For the Power diagram generation, the DMG- α library is used, see [35].

This approach enables an uneven distribution of Voronoi cell sizes that approximates a Fuller curve from $l_{\min}$ to $l_{\max}$, as illustrated in Figure 4a.

The model geometry is created by randomly sequentially placing nodes into the volume domain of the intended model, Ω . We restrict the mutual distance between nodes, d , to lie within the range $d_{\min} \leq d \leq d_{\max}$. The generator starts with the maximum interparticle distance $d = d_{\max}$ and randomly places nodes into the domain, ensuring that no two nodes are closer than d . After 10^5 consecutive failed placement attempts, the algorithm decreases d to the next smaller value on the Fuller curve and repeats the process. The generation ends upon reaching a target domain saturation, S_{target} , typically set to 95%.

To create a heterogeneous mesostructure resembling generic concrete, we shrink the cells toward their respective centers of gravity, creating intermediate zones that will eventually contain the cement matrix (see Figure 4b). The minimal distance of generating nodes and their radii during initial node placement reflect this shrinking so that the final grains have the intended size. Depending on the expected level of aggregate compaction, the shrinkage

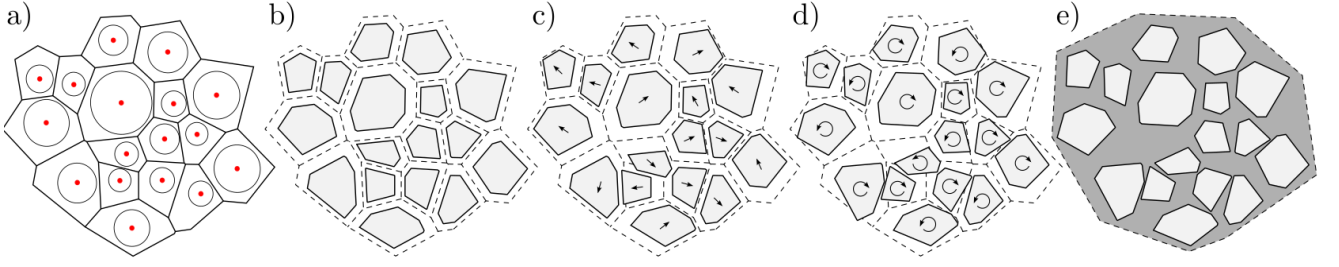


Figure 4: An illustration of the mesostructure generation process: a) Power diagram of the input nodes, b) shrinking the cells (aggregates) towards their center of gravity to create the matrix volume, c) a random translation of the cells, d) random rotation of the cells, e) the resulting mesostructure with aggregates and cement matrix.

may vary between 10% and 30% of the cell volume.

To further enrich the structure, each cell is randomly shifted within the available surrounding space (see Figure 4c). Finally, to break the mutual parallelism of cell surfaces, each cell is rotated by a random 3D angle (see Figure 4d). During this process, we ensure that no two cells (grains) collide. In this way, a mesostructure resembling that of concrete is created; compare Figure 4e with Figure 3. A 3D illustration of the created structure is shown in Figures 6b and 6c.

For collision detection between grains, the Separating Axis Theorem (SAT) [36] is applied. The core idea of SAT is to identify if there exists an axis along which the projections of two polyhedra do not overlap. If such an axis is found, the polyhedra are not colliding; if no separating axis exists, a collision is detected.

The method begins by calculating the normal vectors (axes) for the faces of each polyhedron. These normals serve as potential separating axes. The vertices of the polyhedra are then projected onto these axes. If, for any axis, the projections do not overlap, the algorithm returns False, indicating no collision. If the projections overlap on all axes, the algorithm returns True, indicating a collision. This approach is efficient and widely used for detecting collisions between convex polyhedra. A schematic algorithm of the described mesostructure generation process is provided in Algorithm 1.

Algorithm 1 Random Sequential Node Placement and Mesostructure Generation

Input: Domain Ω , distance range $[d_{\min}, d_{\max}]$, target saturation S_{target}

Initialize $d \leftarrow d_{\max}$

Initialize node set $N \leftarrow \emptyset$

while Saturation $S < S_{\text{target}}$ **do**

failures $\leftarrow 0$

while failures $< 10^5$ **do**

Randomly generate node $n \in \Omega$

if $\forall m \in N, \|n - m\| \geq d$ **then**

Add n to N

Update saturation S

Reset failures $\leftarrow 0$

else

Increment failures

end if

end while

Decrease d to the next value on the Fuller curve

end while

Step 2: Shrink Cells

for each cell C_i in the structure **do**

Compute center of gravity G_i

Shrink C_i towards G_i by a factor $s \in [10\%, 30\%]$

end for

Step 3: Random Cell Shifting and Rotation

for each cell C_i **do**

while C_i collides **do**

Randomly shift C_i within allowed bounds

Rotate C_i by a random 3D angle as a rigid body

Collision Detection Using SAT

for each already placed cell C_j **do**

Face normals as potential separating axes

Project vertices of C_i and C_j onto these axes

if any axis separates the projections **then**

No collision detected

Add C_i to the list of placed cells

else

Collision detected

Reattempt random shift and rotation for C_i

end if

end for

end while

end for

Output: Generated mesostructure geometry

3.2. On aggregate granularity

In general, Voronoi polyhedra resemble the shape of real aggregates well, however it is difficult to match the desired sieve curve. During the generation of nodes for the weighted power tessellation, distribution of weights (radii) is defined exactly according to a Fuller curve:

$$P(D) = 100 \left(\frac{D}{D_{\max}} \right)^n \quad (3)$$

where $P(D)$ is the percentage passing for particle diameter D , and $D_{\max}$ is the maximum particle diameter and n is the Fuller exponent (typically 0.5). For restrained granularity, e.g. 4–16 mm, a modified Fuller curve reads:

$$P(D) = 100 \frac{D^n - D_{\min}^n}{D_{\max}^n - D_{\min}^n} \quad (4)$$

where $D_{\min}$ and $D_{\max}$ are the minimum and maximum particle diameters, respectively. The theoretical Fuller curve as described in Eq. 4 is plotted red in the plots of Figure 5. The Fuller curve used for generation of Power tessellation radii is plotted green along with their relative histogram.

After the initial nodes are generated, the Voronoi polyhedra resulting from the tessellation no longer match the Fuller curve, and this discrepancy increases after the subsequent shrinking of the cells (see Figure 4) used for modeling the mesostructure of concrete. The Fuller curve obtained from the actual resulting 3D aggregates (more precisely the diameters of the spheres of equivalent volume) is plotted in blue along with their relative histogram. It can be seen in Figure 5a that shrinking the cells by, for example, 30% of their absolute dimension towards their center of gravity results in a significant deviation from the intended Fuller curve. To better approximate the Fuller curve on average, one might consider increasing the limiting diameters for the Power tessellation, as shown in Figure 5b. Indeed, this approach yields a closer overall match.

If an even better match is desired, the cells can be shrunk more carefully by weighting their respective shrinking ratios (initially e.g., 30%) according to their 3D volume percentile, p_i , among other cells. For instance, good results are achieved by applying a probability density function of a symmetrical Beta distribution with parameters $\alpha = \beta = 1.8$. The weighted shrinking coefficient for cell i then reads:

$$s_c(p_i) = 0.3 \cdot \frac{p_i^{\alpha-1}(1-p_i)^{\beta-1}}{B(\alpha, \beta)} \quad (5)$$

The weighting factor is given by the probability density function of the Beta distribution with shape parameters α and β , evaluated at the cell's volume percentile p_i . The function $B(\alpha, \beta)$ is the Beta function, defined as

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1} dt$$

which serves as a normalization constant to ensure that the total area under the Beta distribution probability density function equals 1. This normalization is essential for $s_c(p_i)$ to represent a proper weighting factor. Figure 5c shows that this modification leads to a solid match between the Sieve curve of the resulting 3D grains and the desired ideal Fuller curve.

In case that matching a different sieve curve is desired, the Power tessellation radii can be generated exactly as desired and a similar iterative approach of a weighted shrinking can be applied.

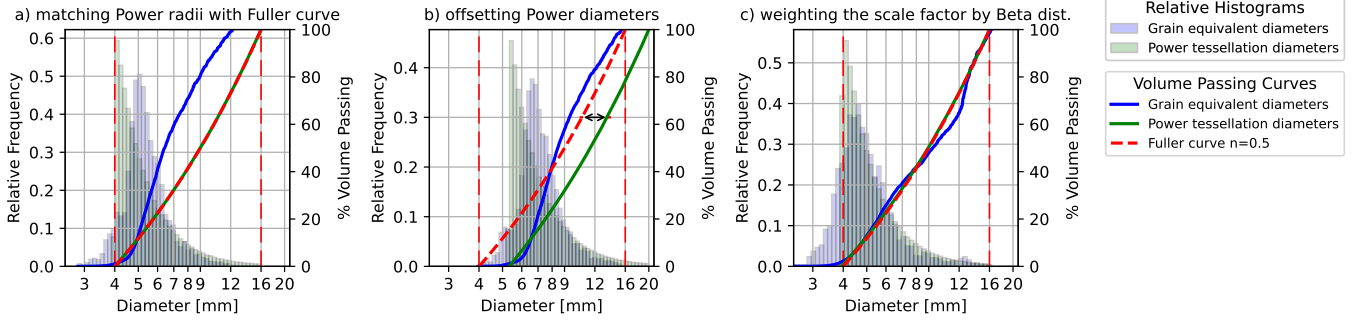


Figure 5: Comparison of sieve curves derived from the theoretical Fuller distribution (red), the input radii for Power tessellation (green), and the actual 3D aggregate diameters after tessellation and shrinking (blue). (a) Uniform 30% shrinking of Voronoi cells leads to a significant deviation from the target distribution. (b) Adjusting the tessellation input range reduces the mismatch. (c) Applying volume-percentile-weighted shrinking using e.g. a Beta distribution ($\alpha = \beta = 1.8$) yields a close match to the intended Fuller curve.

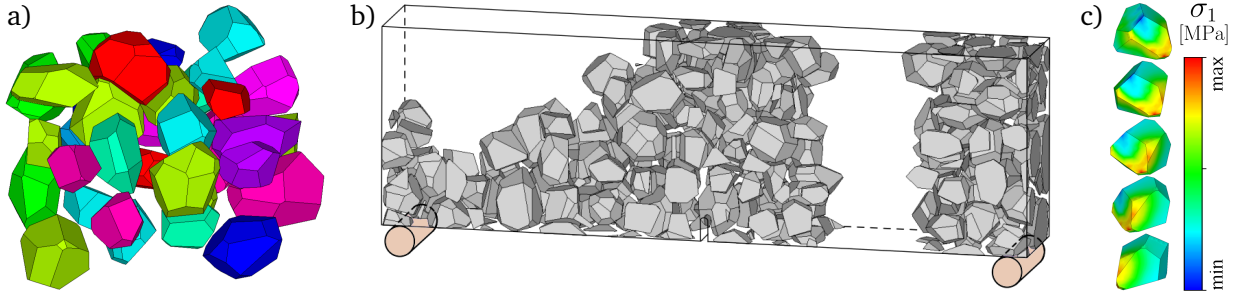


Figure 6: a) A 3D representation of aggregates, b) example of the mesostructure within a notched three-point bending specimen, c) exemplary σ_1 stress field on the surface of an aggregate.

3.3. Meshing and preprocessing

The described mesostructure can be generated to fill the exact volume of the intended numerical model, or alternatively, the model can be extracted (cut) from a larger bulk material structure. This choice depends on whether the mesostructure near the model boundaries should reflect influences such as formwork or a specimen cut from the bulk material.

The resulting geometry and topology of the aggregate structure are imported into the ANSYS framework using the PyMAPDL library [37], where volume objects are created in accordance with the topology. For complex 3D topologies, mesh generation can constitute a significant portion of the total computational cost. The default ANSYS APDL mesher was found to be insufficient in terms of efficiency and performance. Therefore, we chose to employ the highly efficient Gmsh mesh generator [38], which offers significant improvements in both speed and flexibility. The use of the efficient Gmsh library provides a substantial preprocessing speedup compared to the proprietary ANSYS mesher.

For volumetric meshing of the model, we use 20-node hexahedral SOLID186 elements (suitable for elastic analysis) that are degenerated into their tetrahedral forms. This approach is necessary to effectively mesh geometries with complex features, providing a balance between accuracy and computational feasibility. However, it is critical to carefully select the element size to avoid excessively large stress gradients within the model.

To optimize computational efficiency, regions of the model with expected low stress gradients are coarsened, see Figure 7. These areas exhibit minimal variations in stress distribution, allowing for mesh reduction without significant loss of accuracy. Such an approach was verified e.g. in [39].

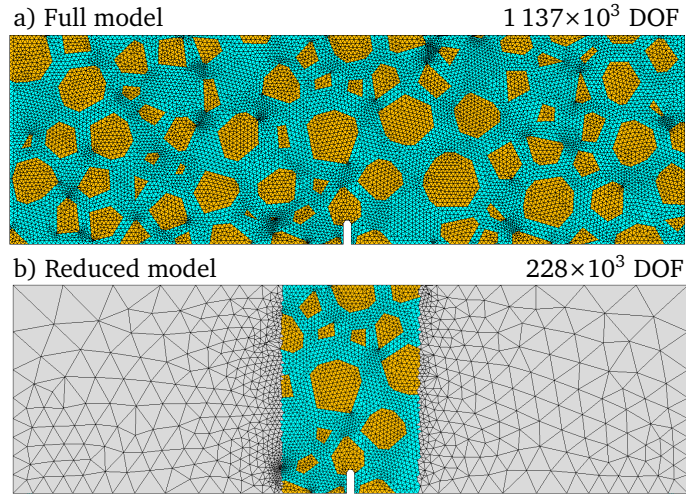


Figure 7: Reducing model complexity by coarsening regions with expected low stress gradients

In this work, without any noticeable loss of precision, computational demands were reduced by modeling and finely meshing the mesostructure only within the inner 25% of the model width. In this fine-mesh region, the maximum element size was set to 2 mm (see the mesh study in Appendix A). An example ANSYS-Gmsh model generating snippet is provided in Appendix B.

Within the left and right outer 37% of the model width, a coarser mesh was applied, representing a homogenized material with properties averaged between the cement matrix and aggregates. In this coarse region, the maximum element size was generally unrestricted, except for the support regions where it was limited to 6 mm to ensure adequate resolution.

Subsequently, the assembled numerical models were executed on the computational resources of the Czech IT4Innovations National Supercomputing Center. This setup benefits from significant solution speedup due to the utilization of multiple multiprocessor computational nodes with extensive memory and storage capabilities. The complete modeling pipeline is shown in Figure 8. It summarizes all steps from geometry generation to final analysis.

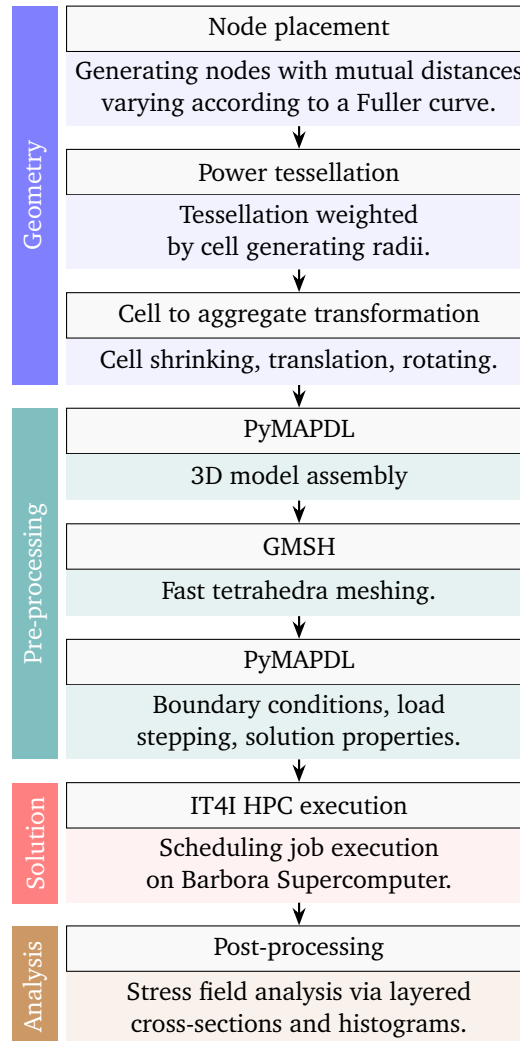


Figure 8: Stages of the MFEM modeling pipeline.

Table 1 presents a comparison between the full and reduced finite element models in terms of computational efficiency, resource requirements, and energy consumption. The table lists the number of volume elements and degrees of freedom (DOFs) for each model, as well as the meshing time in Gmsh, the ANSYS solution time using 8 computation nodes (288 cores of Intel Cascade Lake 6240 CPU), the RAM required for the in-core solution of the FEM problem, and the total energy consumed during computation.

The full model, shown in Figure 7a, consists of approximately 2.04 million volume elements and 1.137 million DOFs. It requires 15 minutes for meshing, 148 GB of RAM, and 16 minutes for the solution, consuming 0.597 kWh of energy. In contrast, the reduced model, shown in Figure 7b, contains 406 thousand volume elements and 228 thousand DOFs, significantly reducing computational cost with a meshing time of 90 seconds, a RAM requirement of 41 GB, and an ANSYS solution time of only 3 minutes, consuming 0.107 kWh.

Table 1: FEM Model Comparison

	Full model	Reduced model
Volume Elements	$2\,040 \times 10^3$	406×10^3
Degrees of Freedom	$1\,137 \times 10^3$	228×10^3
Gmsh Meshing	15 min	90 sec
ANSYS Solution	10 min	1.5 min
RAM Required	148 GB	41 GB
Energy Used	0.597 kWh	0.107 kWh
Energy Cost	0.15 EUR	0.03 EUR

Figure 9 illustrates the trade-offs between execution time, energy consumption, and scalability. In subfigure (a), execution time decreases significantly with increasing CPU cores, but the performance gain flattens beyond 72–108 cores, while energy consumption rises steadily. Subfigure (b) confirms this: stepwise speedup drops sharply with each additional node, while the energy increase ratio remains above 1 and continues to grow. All cores were utilized above 99% throughout execution, ruling out idle hardware as a bottleneck. Each group of 36 cores corresponds to a computational node (two 18-core CPUs), so scaling introduces both intra-node and inter-node communication overheads, which progressively degrade efficiency.

Beyond 2–3 nodes, the marginal performance gain is minimal, and the energy cost continues to rise. Moreover, the HPC system is governed by SLURM-based scheduling [40], where users typically wait longer in the queue to receive allocations of more nodes—even for short runs—than for allocations of fewer nodes with longer runtimes. This makes obtaining large node allocations more time-consuming and challenging in practice. In this context, it is not only more energy-efficient but also operationally advantageous to run multiple models in parallel on fewer nodes each. This strategy improves overall solution throughput by better matching available resources and queue dynamics, rather than saturating the system with a single, inefficiently scaled job.

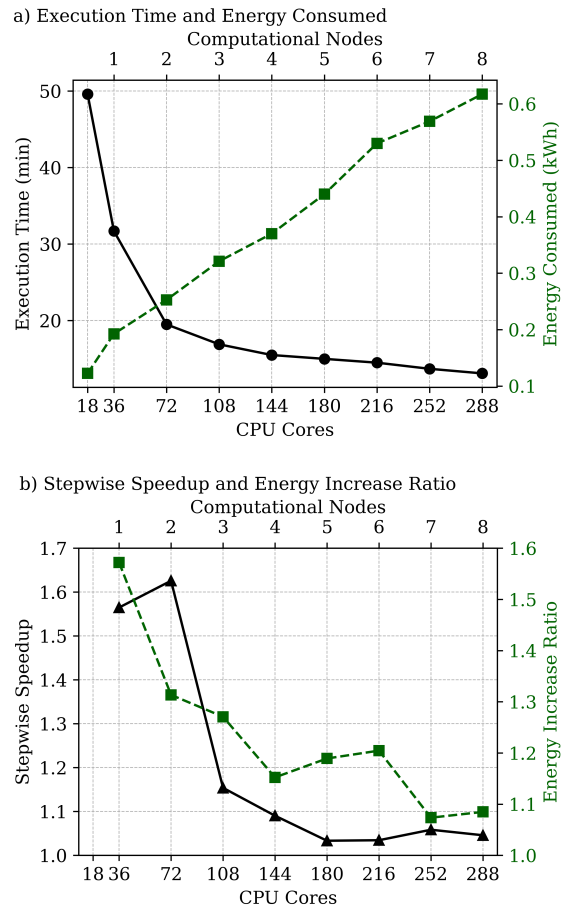


Figure 9: Parallelization efficiency: a) Execution time and energy consumed, b) stepwise solution speedup and stepwise energy increase ratio.

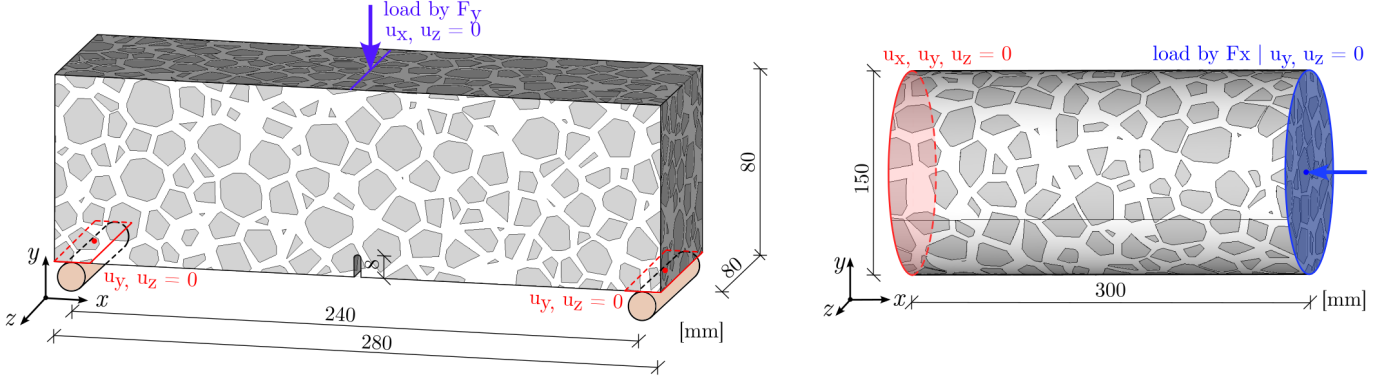


Figure 10: Boundary conditions and dimensions numerical models: three-point bending specimen (unnotched in Sec. 4.1, notched in Sec. 4.2) and compressed cylinder (Sec. 4.3).

4. Numerical models

In the text below, we present the geometries of the studied numerical models, including boundary conditions and analytical solutions of stress field distribution within several common concrete test specimen setups: an unnotched (blunt) three-point bending (Section 4.1), a notched three-point bending (Section 4.2), and a compressed cylinder (Section 4.3).

4.1. Blunt three-point bending setup

The first model is a standard unnotched (blunt) three-point bending specimen. The specimen is illustrated in Figure 10 (consider it in the unnotched variant). The test specimen has a span $S = 240$ mm, depth $D = 80$ mm, thickness $T = 80$ mm.

The boundary conditions for the supports are modeled by creating rigid regions on the beam's bottom face (CERIG elements). Each rigid region is associated with a master node (MASS21), located at the region's center of gravity, shown in red in Figure 10. The distance between the two master support nodes corresponds exactly to the beam span S . At the supports, zero displacements are imposed in the y - and z -directions (denoted $u_y = u_z = 0$), along with zero rotation about the X -axis and zero overall rotation of the model about the Y -axis. The loading master node is assigned a vertical force $F_y = 1$ N to represent a unit load; its horizontal displacements u_x and u_z are constrained to zero.

For the blunt 3PBT geometry, the uniaxial normal stress component σ_x is examined, as it can be directly compared to the corresponding normal stress from Euler-Bernoulli beam theory. This comparison takes into account the statistical nature of the mesostructure, as discussed in Section 5.1.

The analytical solution for the normal stress distribution in a linear homogeneous material is:

$$\sigma_{x,LM} = \frac{FL}{4I}y, \quad (6)$$

where F is the applied force (unitary here), L is the beam span, I is the moment of inertia of the beam cross-section, and y is the vertical distance from the neutral axis.

However, relying solely on normal stress for damage assessment can be misleading, as fracture or damage initiation often occurs under multiaxial stress states, especially when the concrete mesostructure is explicitly modeled.

4.2. Notched three-point bending setup

A standard three-point bending test setup is used, encompassing both notched and unnotched variants. The test specimen has a span $S = 240$ mm, depth $D = 80$ mm, thickness $T = 80$ mm, and, for the notched variant, a notch height $a = 0.1D$. The notch width is 3 mm and its tip is rounded, as illustrated in Figure 10. It is important to note that the stress values $\sigma_{x,LEFM}$ are computed assuming a sharp notch tip, so strict numerical comparison with the rounded-tip model cannot be enforced.

The boundary conditions for the top load are modeled similarly to the blunt case. A narrow rigid region is created across the top surface of the model, with a master node assigned at its center. The loading master node is assigned a vertical force $F_y = 1$ N representing a unit load, while its horizontal displacements u_x and u_z are constrained to zero.

The normal stress distribution within the 3D FEM model is compared to the response of a linear homogeneous material obtained from a 2D FEM counterpart model under plane strain conditions. This 2D model was employed to extract bending LEFM stress σ_{xx} along the specimen height W . To accurately capture the crack-tip singularity, the notch tip was meshed using the KSCON command, which collapses quadrilateral elements into triangular ones and moves mid-side nodes to one-quarter of the element length. The 2D mesh consisted of 8-node quadrilateral elements of type PLANE183. The 2D model dimensions, loading, and material parameters corresponded to those of the 3D mesoscale FEM model. The 2D numerical solution was validated against the analytical stress intensity factor K_I from [41], achieving an error below 1% [42]. After confirming the good agreement of K_I , the bending stress σ_{xx} was extracted for comparison. The results demonstrate that the most frequent stress values closely align with the analytical solution, with only minor low-frequency noise around these values.

4.3. Compression-loaded cylinder

The second test setup is a clamped compression-loaded cylinder with height $H = 300$ mm and diameter $D = 150$ mm. Rigid regions of nodes are created on both cylinder faces and assigned to their respective master nodes located at the geometric centers, as shown in Figure 10. The master node on the loaded (top) face (blue) is constrained in all rotations and displacements except for the vertical displacement load u_y . The master node on the supported (bottom) face (red) is fully restrained in all translations and rotations.

The normal stress distribution within the proposed FEM model is compared to the analytical solution for a linear homogeneous material:

$$\sigma_{x,LM} = \frac{N}{A}, \quad (7)$$

where N is the applied axial force (here unitary), and A is the cylinder cross-sectional area.

For a homogeneous material, the numerical model reproduces the expected mechanical response in terms of normal stress distribution as predicted by linear elastic fracture mechanics. Experimental studies reporting stress fluctuations under compression can be found in [43, 44, 45, 46].

5. Numerical results and discussion

The sections below detail the methodology used for numerical data analysis of the studied geometries. Special attention is given to the extraction of stress data from the simulation results for each geometry. We then present the local stress distribution within aggregates derived from the internal stress state. This is followed by an in-depth analysis of the influence of stiffness ratio on the stress field, including a statistical evaluation of this complex problem.

5.1. Numerical data analysis

To analyze the stress field in the models, we extract stress values from multiple parallel cross-sections along the specimen. The cross-sections are positioned sufficiently close relative to the specimen's scale to cover volumes experiencing similar mechanical load states. This allows us to treat the extracted stress values as representative of a single cross-section of finite thickness. For it might prove useful for a future usage, Listing 1 presents the APDL script employed to extract the specified stress values and node coordinates for each cross-section plane.

```
! Offsets the working plane to specific x-coordinate
WPOFFS,0.140,,
! Rotates the working plane by 90 degrees about third rotation about Y axis so the XY WP plane
  becomes the global YZ plane
WPROTA,,,90
! Select elements by material
ESEL,S,MAT,,MATERIAL
! Create a surface called surfName. Refinement level 2
SUCR,SURFNAME,CPLANE,2
! Map results onto selected surface
SUMAP,RN,S,X
! Plot result data on selected surface
SUPL,SURFNAME,RN,0
! Moves surface geometry and mapped results to an array parameter
SUGET,SURFNAME,RN,CSECTION,1
! Retrieves array dimensions from the csection array and stores them into scalar parameters
*GET, ARRSIZE1, PARM, CSECTION, DIM,1
*GET, ARRSIZE2, PARM, CSECTION, DIM,2
*GET, ARRSIZE3, PARM, CSECTION, DIM,3
! Extracts x, y, z coordinates from csection array into a new array surf_OUT with dimensions
  arrSize1 by (arrSize2 - 4)
*DIM,SURF_OUT,ARRAY,ARRSIZE1,ARRSIZE2-4
*VFUN,SURF_OUT(1,1),COPY,CSECTION(1,1)
*VFUN,SURF_OUT(1,2),COPY,CSECTION(1,2)
*VFUN,SURF_OUT(1,3),COPY,CSECTION(1,3)
! Copy eighth component (SX stress) from csection array to surf_OUT
*VFUN,SURF_OUT(1,4),COPY,CSECTION(1,8)
! Write the surf_OUT array data to text file named by %FILENAME% at path %PATH1%
*MWRITE,SURF_OUT(1,1,1),'%FILENAME%',TXT,'%PATH1%'
! Format specifier for output file: 7 exponential numbers per line with width 20, 8 decimals
(7E20.8)
```

Listing 1: ANSYS APDL Script: Surface stress extraction

Without significant loss of information, we further condense the data by dividing the cross-section height into many thin layers, over which histograms of stress values are constructed. By assembling these histograms at their respective vertical positions, we obtain a comprehensive summary of the stress distribution across the investigated volume of the model. This approach is illustrated in Figure 11 for a three-point bending test specimen. Figure 12 shows an example of the normal stress field across such a cross-section, highlighting differences between aggregate and matrix phases with unequal Young's moduli.

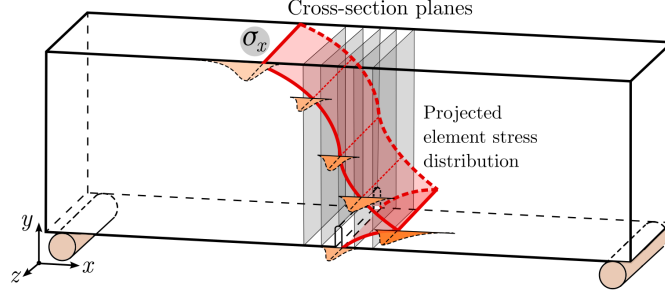


Figure 11: Cross-sectional planes used for statistical data analysis.

By this evaluation method, no information is lost or averaged out; the visualization preserves details on stress extremes as well as the full distribution of stresses.

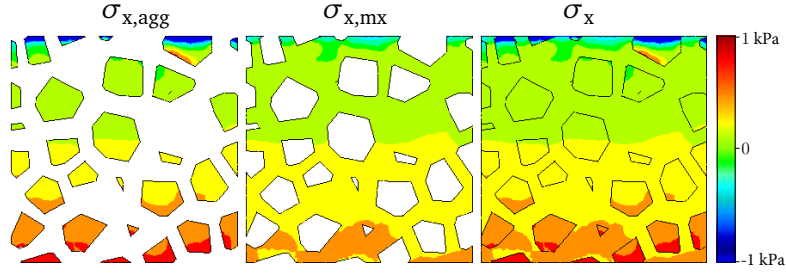


Figure 12: Normal stress across a 2D cross-section within the aggregate material ($\sigma_{x,agg}$), matrix material ($\sigma_{x,mx}$) and full cross-section (σ_x).

The results presented below were acquired within 10 equidistant cross-sections spread across the notch width, extracted from 50 numerical models with unique aggregate structures.

5.2. Influence of Stiffness Contrast on Local Stress Distribution

To assess the effect of mesoscale stiffness heterogeneity on the internal stress state of concrete, we conducted an elastic finite element analysis focusing on a single polyhedral aggregate embedded within a continuous cementitious matrix. The first principal stress, σ_1 , was evaluated under uniaxial loading conditions for varying stiffness ratios defined as $R_{M/A} = E_{mx}/E_{agg}$, ranging from 0.1 to 10. As illustrated in Figures 13 and 14, the stress field within the inclusion is highly sensitive to $R_{M/A}$.

For softer aggregates ($R_{M/A} > 1$), the aggregate remains nearly stress-free, with the matrix accommodating the majority of the external load. Conversely, for softer matrix ($R_{M/A} < 1$), the aggregate attracts and concentrates both tensile and compressive stresses, with pronounced localization of stress near sharp geometric features such as corners and edges, see Figure 13 for σ_1 and Figure 14 for σ_3 principal stress, respectively.

This behavior aligns qualitatively with classical inclusion theory—specifically Eshelby's solution for ellipsoidal inclusions in infinite media [47]. According to Eshelby, a homogeneous ellipsoidal inclusion embedded in a linearly elastic infinite matrix under uniform remote stress experiences a uniform internal stress and strain field, which

can be expressed analytically. However, our case involves a faceted (polyhedral) inclusion with non-ellipsoidal geometry, where Eshelby's assumptions no longer hold. In such cases, the internal stress field becomes non-uniform and exhibits singularities at geometric discontinuities. Analytical and numerical extensions, such as those presented in [48], confirm that polyhedral inclusions produce more complex stress fields, with stress intensification driven both by stiffness mismatch and inclusion geometry.

These findings highlight the pivotal role of stiffness contrast and morphological features in governing local stress distributions, which are central to the initiation of microcracks and the subsequent fracture evolution in quasi-brittle heterogeneous materials like concrete.

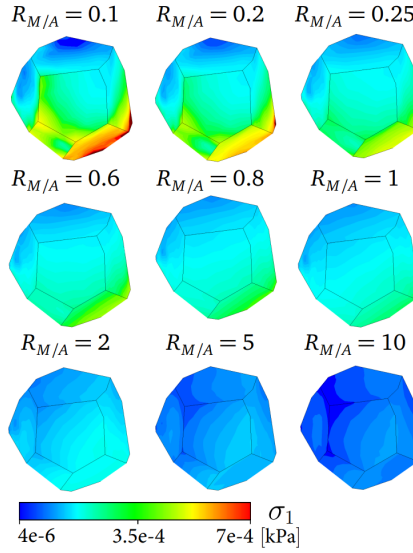


Figure 13: Development of the principal stress σ_1 distribution on a grain surface as influenced by stiffness heterogeneity.

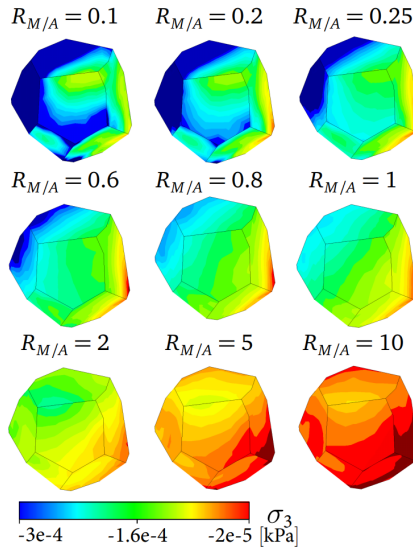


Figure 14: Development of the principal stress σ_3 distribution on a grain surface as influenced by stiffness heterogeneity.

5.3. Stress field dependence on aggregate to matrix stiffness ratio

In what follows, we investigate the influence of increasing stiffness contrast between the aggregate and matrix material phases. The aim is to provide insight into how the stress distribution within the material evolves as the heterogeneity of material properties between the matrix and aggregate phases increases.

This investigation is particularly relevant for the utilization of alternative aggregate materials, such as recycled crushed bricks (see Section 2). The presence of inclusions significantly softer (lower Young’s modulus) than granite, or even softer than the cement matrix, can lead to substantial increases in stress concentrations within the composite. Such elevated stress concentrations are expected to cause earlier crack initiation during the service life compared to traditional granite aggregates. We believe that these numerical results contribute to a better understanding of concrete’s inherent heterogeneity and the resulting local stress concentrations.

We analyze the evolution of normal stress distribution differences between a homogeneous material and increasingly heterogeneous composites.

As noted in the introduction, the elastic modulus of aggregates can vary widely—from about 3.5 GPa (e.g., crushed bricks) up to 75 GPa or more (e.g., crushed granite). The cement matrix modulus typically ranges between 10 GPa and 30 GPa [49]. This results in stiffness ratios $R_{M/A}$ spanning roughly from 0.1 to 10. More elastic properties of various materials can be found [50]. For this study, the Poisson’s ratio is fixed at $\nu = 0.2$ for both materials. The aggregate structure is generated to resemble a realistic distribution of aggregate sizes between 4 and 16 mm.

The MFEM stress values are displayed as relative histograms across a fine array of layers along the specimen height. In each layer, dark red indicates the most frequent stress values, with frequency decreasing towards dark blue. For homogeneous models ($R_{M/A} = 1$), the most frequent stress values closely match the analytical reference solution, with scattered noise of negligible frequency around these values. Only near the top surface of the TPB models do compressive stress concentrations appear due to the loading boundary condition.

When stiffnesses are comparable ($R_{M/A} \approx 1$), strains in matrix and aggregates are compatible, resulting in a smooth stress distribution. However, as the heterogeneity becomes more pronounced in either direction, the normal stress distributions within matrix and aggregates diverge.

For a softer matrix relative to aggregates ($R_{M/A} < 1$), stress peaks within the matrix tend to smear out because the matrix deforms more easily while transferring load between aggregates. Nonetheless, the matrix remains responsible for overall equilibrium, so its stress cannot approach zero.

Conversely, for softer aggregates relative to the matrix ($R_{M/A} > 1$), the stiffer matrix attracts more stress due to its higher deformation resistance. As aggregate stiffness decreases, they lose the ability to effectively transfer stress across the composite, forcing the matrix to carry a larger portion of the load. In the limiting case, the composite behaves like a cement matrix with void inclusions.

Figures 15 and 16 show the stress distribution in the blunt three-point bending (TPB) model within the matrix and aggregates, respectively. Similarly, Figures 17 and 18 present the stress distribution for the notched TPB model, again distinguishing between matrix and aggregates. Finally, Figures 19 and 20 depict the stress distribution in the compression-loaded cylinder model within matrix and aggregates, respectively.

In all cases, stress values are visualized using a color scale normalized separately for each horizontal layer along specimen height. The scale spans from 0 (blue, least frequent) to 1 (red, most frequent), representing the relative frequency of normal stress within each individual layer. As such, colors indicate local stress distribution patterns

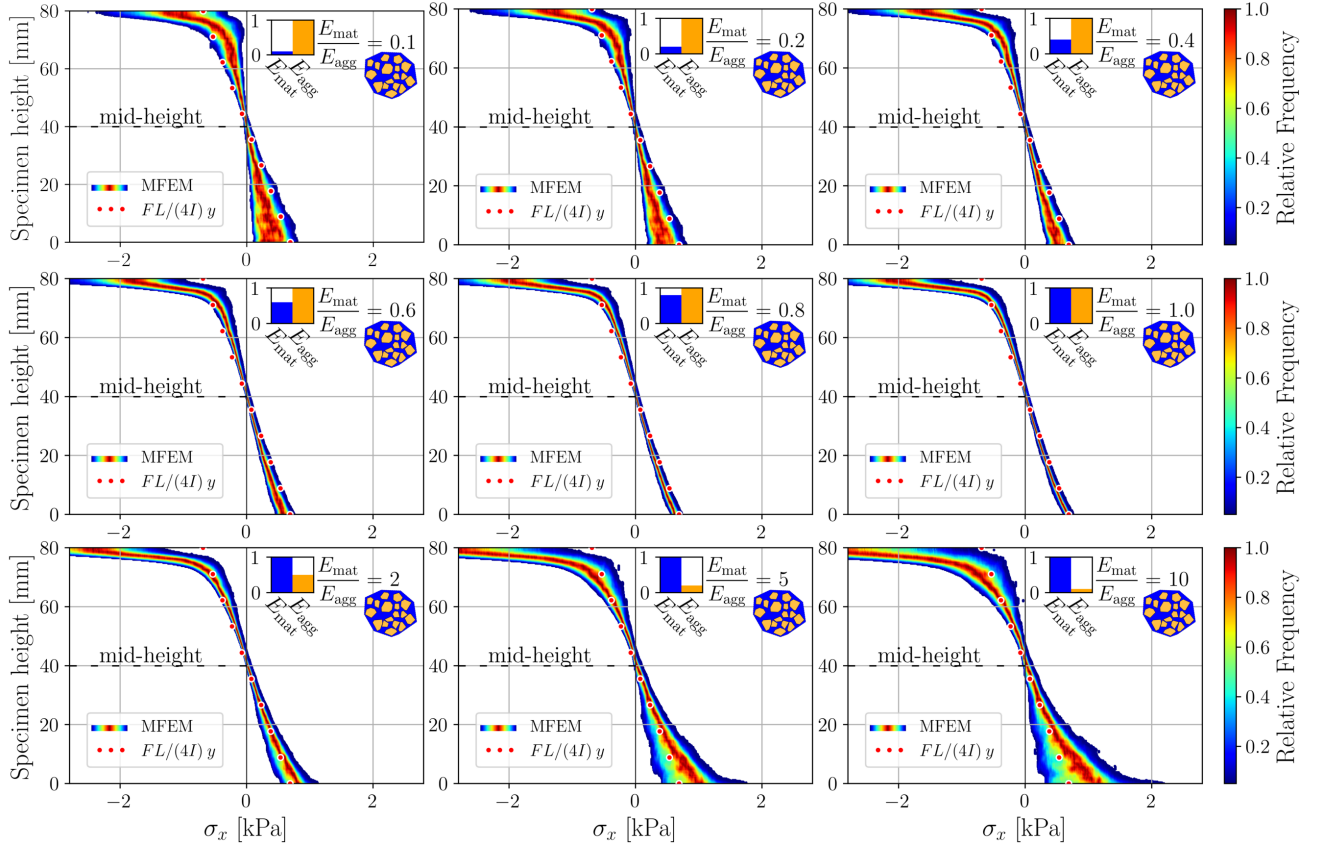


Figure 15: Blunt TPB: matrix normal stress within the notch region in dependence on the heterogeneity parameter, $R_{M/A}$. Colors show relative stress frequency per horizontal layer (red = most frequent).

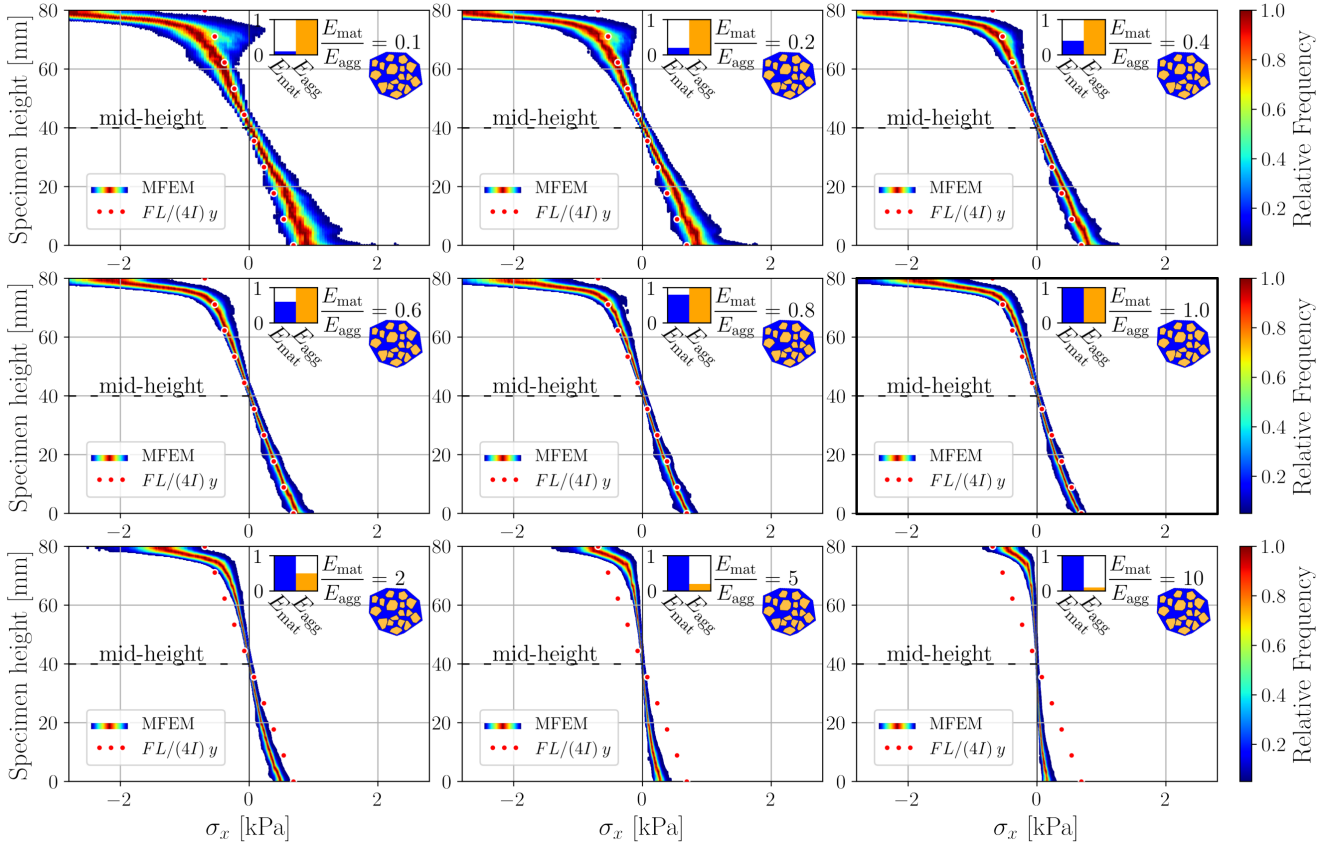


Figure 16: Blunt TPB: aggregates normal stress within the notch region in dependence on the heterogeneity parameter, $R_{M/A}$. Colors show relative stress frequency per horizontal layer (red = most frequent).

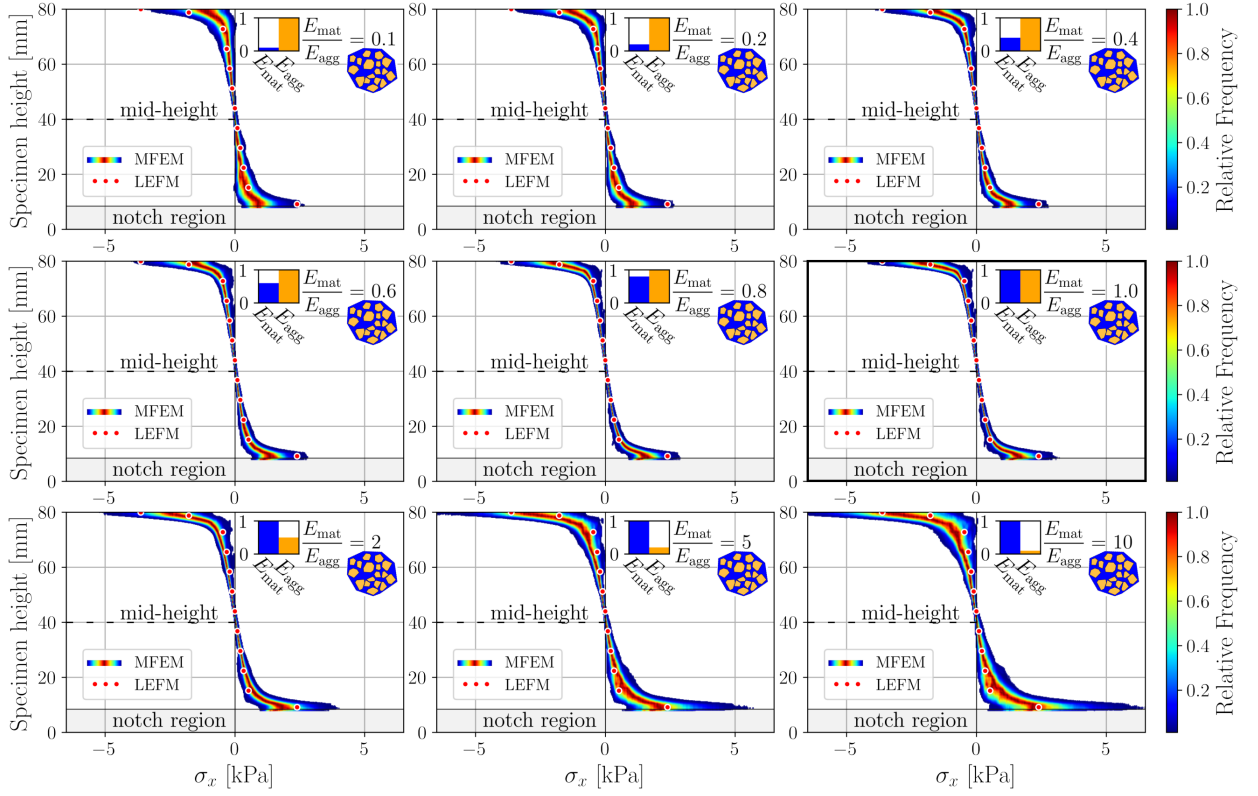


Figure 17: Notched TPB: matrix normal stress within the notch region in dependence on the heterogeneity parameter, $R_{M/A}$. Colors show relative stress frequency per horizontal layer (red = most frequent).

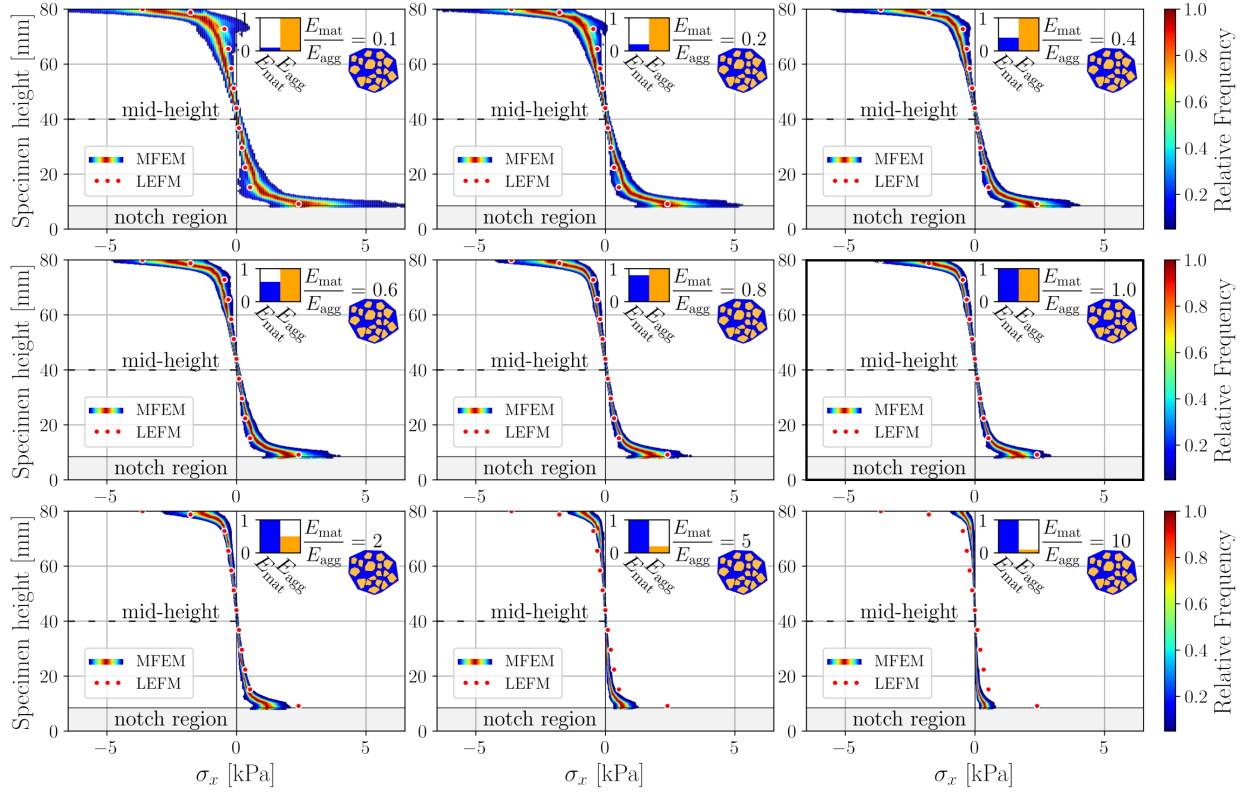


Figure 18: Notched TPB: aggregates normal stress within the notch region in dependence on the heterogeneity parameter, $R_{M/A}$. Colors show relative stress frequency per horizontal layer (red = most frequent).

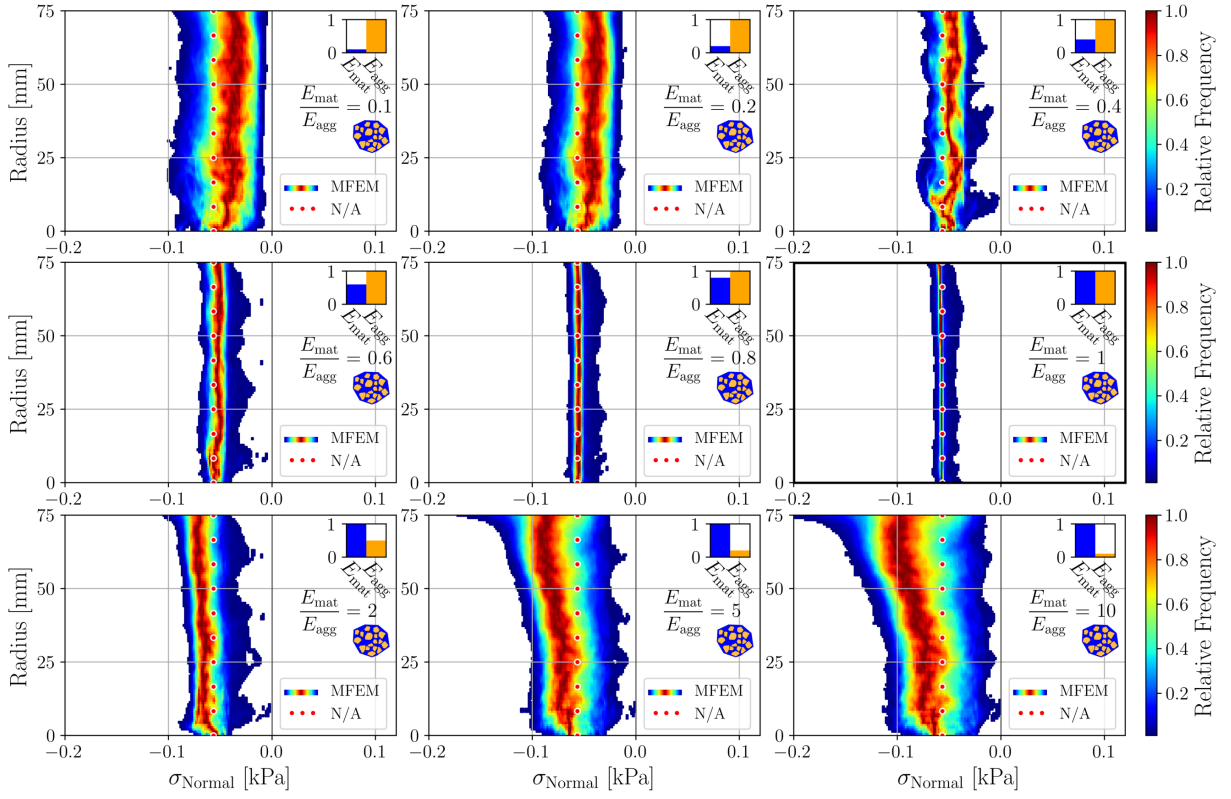


Figure 19: Cylinder compression: matrix normal stress at midspan region in dependence on the heterogeneity parameter, $R_{M/A}$. Colors show relative stress frequency per horizontal layer (red = most frequent).

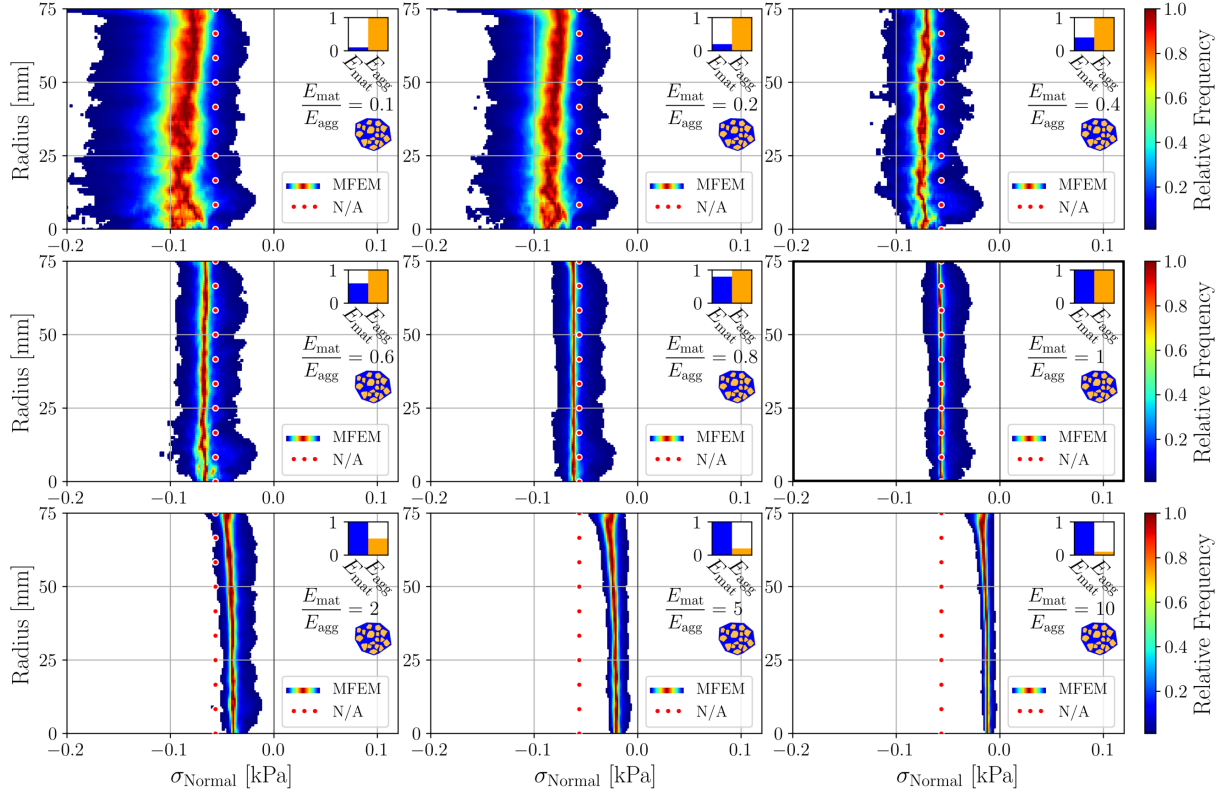


Figure 20: Cylinder compression: aggregates normal stress at midspan region in dependence on the heterogeneity parameter, $R_{M/A}$. Colors show relative stress frequency per horizontal layer (red = most frequent).

This phenomenon is further illustrated in Figure 21, which shows maximum normalized normal stress quantiles, $\bar{\sigma}_{x,\max}(q)$:

$$\bar{\sigma}_{x,\max}(q) = \frac{\sigma_{x,\max,\text{hetero}}(q)}{\sigma_{x,\max,\text{homo}}(q)}, \quad (8)$$

where $\sigma_{x,\max,\text{hetero}}(q)$ is the q th quantile of maximum normal stress in the heterogeneous material and $\sigma_{x,\max,\text{homo}}(q)$ is the corresponding quantile for the homogeneous material ($R_{M/A} = 1$).

The results clearly indicate that the most favorable stress distribution occurs when the stiffness of inclusions closely matches that of the matrix. In this balanced state, stress is more evenly shared between phases, minimizing localized concentrations that may trigger damage. Deviations from this balance—either with significantly softer or stiffer inclusions—lead to pronounced stress redistribution and localization, especially at interfaces and near sharp polyhedral features.

This effect is particularly critical when replacing conventional stiff aggregates, such as granite, with alternative materials for sustainability or cost reasons. The mechanical consequences of high stiffness contrast must be carefully assessed, as they may compromise the composite's structural integrity or durability. Our findings emphasize the importance of considering mesoscale mechanical compatibility—not just material availability—in the selection of aggregates for concrete design.

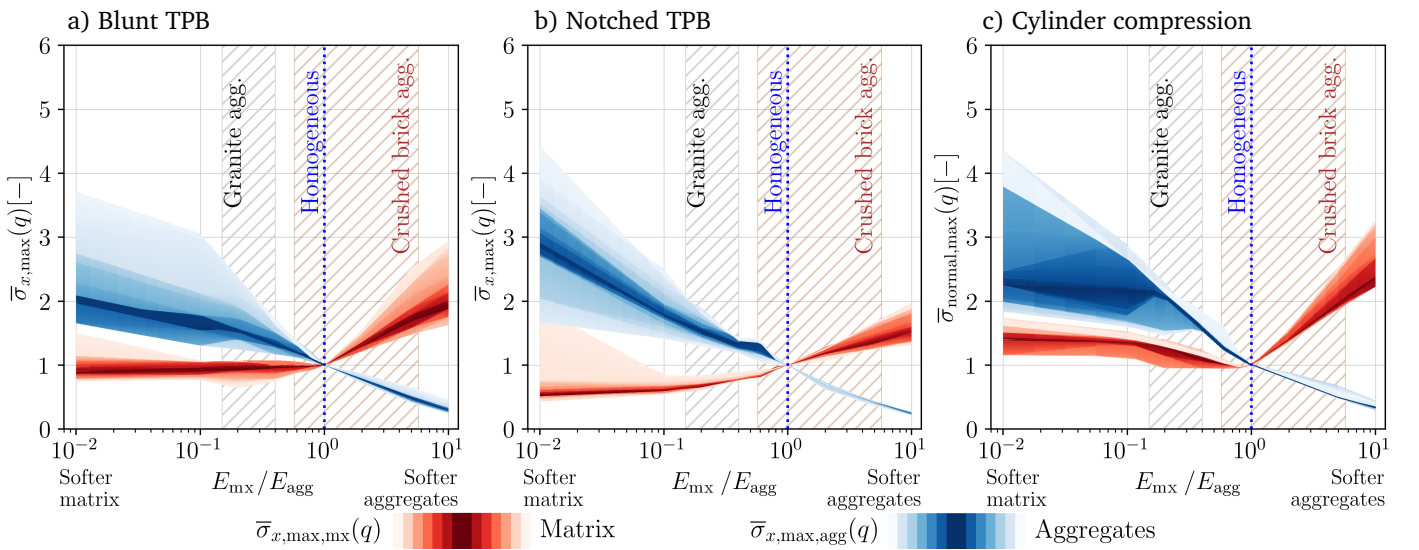


Figure 21: Normed normal stress quantiles $\bar{\sigma}_{\text{Normal,max}}(q)$.

5.4. On aggregate shape influence

To complement the study presented above, we analyze the effect of aggregate shape on stress concentrations within the mesostructure. The geometry of individual aggregates significantly influences the stress distribution in the composite; for a comparison of round versus polyhedral aggregates, see e.g., [51]. As expected, round aggregates produce lower maximal stress peaks than polyhedral ones. Even minimal deviations from roundness in otherwise nearly spherical polyhedra can trigger increased stress concentrations.

In this work, we focus exclusively on polyhedral aggregates representing crushed materials. We investigate the effect of increasing grain angularity (i.e., decreasing sphericity) by modifying the shrinking of tessellation cells (recall Sec. 3.1). For each cell, the principal axes of inertia are determined, and the original shrinking factor s_c (applied uniformly to all vertices toward the cell's center of gravity) is scaled differently along each principal axis.

In Figure 22a, cells are shrunk uniformly, i.e., 100% of s_c along x_1 , x_2 , and x_3 . This configuration represents the most spherical polyhedral aggregates attainable with the present generation procedure, serving as the upper bound of roundness within the constraints of faceted geometry. Figure 22b shows uneven shrinking with $x_1 = 50\% \cdot s_c$, $x_2 = 100\% \cdot s_c$, and $x_3 = 200\% \cdot s_c$, while in Figure 22c, shrinking is more pronounced: $x_1 = 25\% \cdot s_c$, $x_2 = 100\% \cdot s_c$, and $x_3 = 250\% \cdot s_c$. Uneven shrinking generates more angular, less spherical aggregates, as illustrated in the middle panels of Figure 22.

For each convex polyhedral aggregate, sphericity is computed using the standard Wadell formula [52]:

$$\Psi = \frac{\pi^{1/3}(6V)^{2/3}}{A} \quad (9)$$

where V is the enclosed volume and A the total surface area of the aggregate. Sphericity directly quantifies deviation from a spherical form, with $\Psi = 1.0$ corresponding to a perfect sphere; lower values indicate more elongated or angular shapes.

We also examine the angular sharpness of aggregate edges. For each face f , the outward unit normal $\mathbf{n}_f$ is computed. For each pair of faces (f, g) sharing an edge, the dihedral angle is defined as:

$$\theta_{fg} = \arccos(\mathbf{n}_f \cdot \mathbf{n}_g) \quad (10)$$

where $\mathbf{n}_f \cdot \mathbf{n}_g$ is the dot product of the normals. The dihedral angle θ_{fg} characterizes edge sharpness: values near 0° indicate nearly coplanar facets with no perceptible edge, $\approx 90^\circ$ corresponds to orthogonal, blocky edges, and $\approx 180^\circ$ indicates acute ridges with opposing facets.

For convex polyhedral aggregates, there exists an inherent upper bound on achievable sphericity: even in the most rounded tessellation-derived shapes, the faceted geometry prevents Ψ from approaching 1.0. As a result, some degree of nonsphericity and angularity is unavoidable.

Histograms of sphericity and dihedral angles of all aggregates for each aggregate shaping variant are shown in the left panels of Figure 22. Finally, for a constant stiffness ratio ($R_{M/A} = 0.2$), in a similar fashion to the stress results of Section 5.3, the right portion of Figure 22 shows the σ_x stress obtained within midspan aggregates 50 unnotched three-point bending models with each grain shaping variant.

In Figure 23, similarly to Figure 21, we now plot the distribution of normal stress within the aggregates and matrix; $\sigma_{x,\text{agg}}$ and $\sigma_{x,\text{mx}}$, respectively. There, $\sigma_{x,\text{max}}(q)$ is the q th quantile of maximum normal stress in the material. We are now interested in the peaks: as highlighted by the 99th percentile lines from matrix and aggregates. Clearly, less spherical aggregates tend to lead to increased stress peaks. Given the generality of this work across materials with varying stiffness, the shown results rather indicate the general trend for already angular faceted

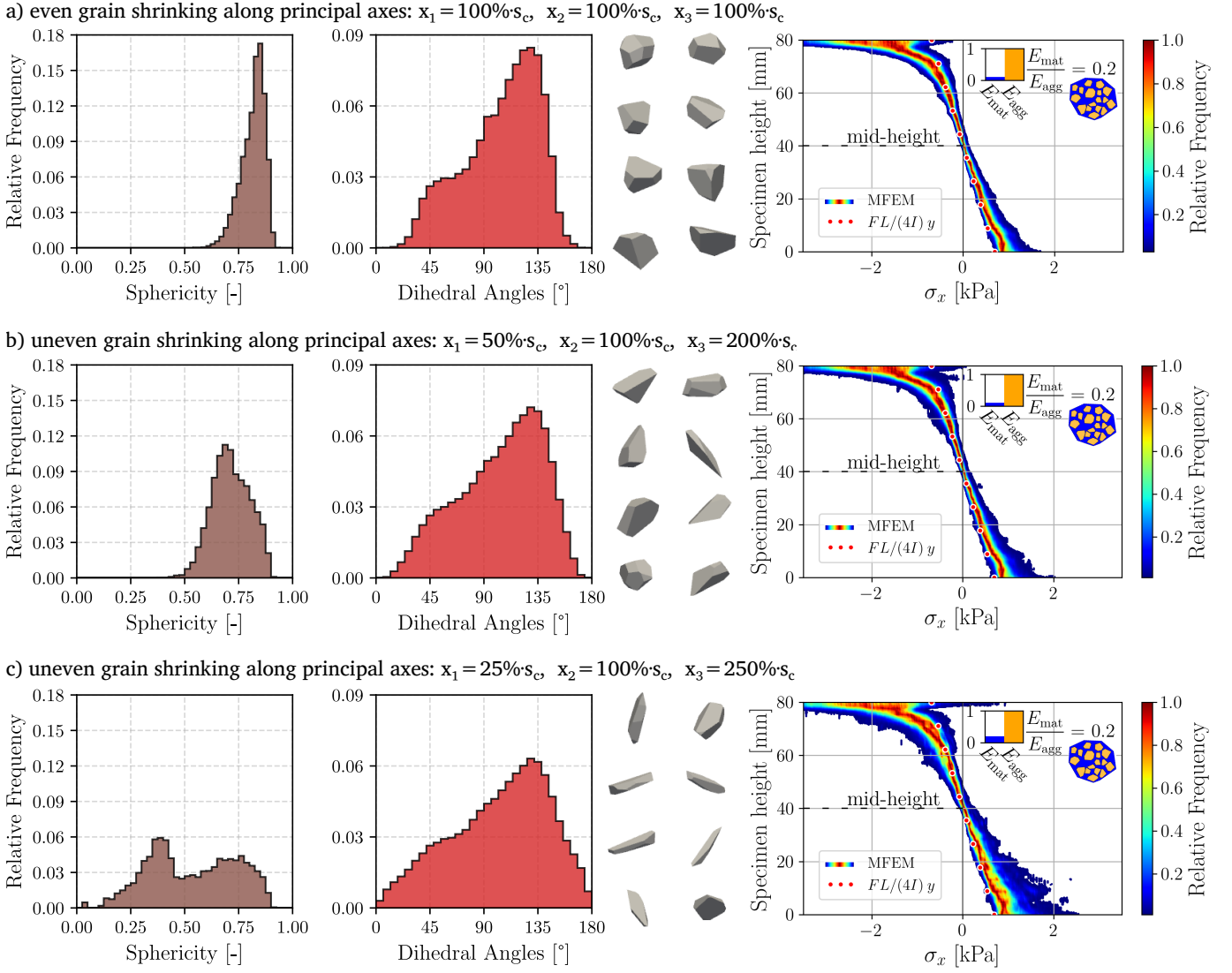


Figure 22: Effect of aggregate shaping on geometry and stress distribution. Left: sphericity Ψ and dihedral angle θ_{fg} distributions. Middle: representative polyhedral aggregates showing increasing angularity from (a) to (c). Right: normal stress σ_x fields, illustrating higher stress peaks with sharper aggregates.

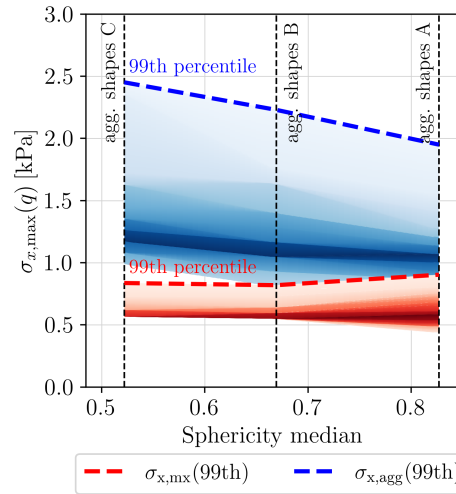


Figure 23: Distribution of maximum normal stress $\sigma_{x,max}$ in aggregates ($\sigma_{x,agg}$) and matrix ($\sigma_{x,mx}$) for varying aggregate angularity. Shown are the 99th percentile peaks, highlighting that less spherical aggregates lead to increased stress concentrations.

aggregates. The actual mechanism behind increasing stress peaks is however hidden by the interactions within each actual mesostructure, governed by mutual aggregate positions and rotations.

6. Conclusions

In the presented work, a Mesoscale Finite Element Model (MFEM) of concrete was developed, enabling detailed representation of its internal heterogeneity. The mesostructure generation pipeline—combining weighted Voronoi tessellation, collision-checked 3D aggregate placement, and high-fidelity meshing—offers a reusable framework for simulating stress distribution and failure mechanisms in a broad class of heterogeneous materials.

Leveraging high-performance computing capabilities, the approach explicitly resolves the aggregate structure, offering transparent control over geometrical and material parameters and offering a more transparent and versatile framework compared to existing models such as LDPM. This makes it suitable not only for concrete, but also for other composites where phase interaction and geometry-induced effects are critical. Unlike black-box approaches such as LDPM, the MFEM setup provides physically interpretable control over input parameters, making it a viable methodology for broader applications, especially when non-traditional or recycled constituents are involved.

The modeling approach is particularly suitable for investigating the influence of aggregate-matrix material contrasts on stress distribution and failure mechanisms—an issue of growing relevance especially when using non-traditional aggregates such as crushed brick recyclate. The numerical study shows that the stiffness mismatch between the phases plays a major role in governing internal stress redistribution and damage evolution and lifetime durability. As the stiffness mismatch increases, two distinct regimes emerge:

- Stiff aggregates in a soft matrix: The compliant matrix undergoes larger deformations, allowing for stress redistribution over a wider region. This tends to reduce local stress concentrations but compromises the overall stiffness of the composite. An excessive matrix deformation may promote early microcracking, especially under fatigue or cyclic loading.
- Soft aggregates in a stiff matrix: The stiff matrix bears the majority of the load, leading to localized stress peaks at matrix-aggregate interfaces. The aggregates no longer contribute much to stress transfer, and may act as stress concentrators rather than reinforcements. This increases the likelihood of crack initiation and propagation, often leading to brittle failure mechanisms. Ultimately, the composite resembles a matrix containing voids, with significantly impaired structural resilience.

In addition to stiffness contrast, the effect of aggregate shape was studied by varying the angularity of polyhedral aggregates. Less spherical, more angular particles were found to produce higher stress peaks, particularly within the aggregate phase, as quantified by sphericity and dihedral angle distributions. This confirms that particle geometry, alongside phase stiffness, plays a significant role in stress localization and the overall stress distribution in the composite.

These observations underline the importance of maintaining a reasonable stiffness contrast between the phases. Ideally, aggregate materials with mechanical properties comparable to those of the matrix ensure efficient load sharing and minimal stress localization. When replacing traditional aggregates like granite with more compliant or recycled materials, such as crushed bricks, the mechanical drawbacks must be carefully assessed to avoid unintended reductions in structural integrity or durability.

Beyond this specific case study, the developed MFEM framework offers a flexible, physically transparent tool for mesoscale stress analysis. A key methodological contribution is the histogram-based statistical post-processing

of stress data across cross-sectional planes, providing quantitative insight into stress concentration distributions. This visualization technique is valuable for interpreting mesoscale mechanical behavior and can aid researchers working on heterogeneous materials modeling.

7. Acknowledgement

This paper was created as part of the project No. CZ.02.01.01/00/22_008/0004631 Materials and technologies for sustainable development within the OP JAK Program financed by the European Union and from the state budget of the Czech Republic. Numerical calculations were done within the IT4Innovations National Supercomputing Center, Czech Republic supported by the Ministry of Education, Youth and Sports of the Czech Republic through the e-INFRA CZ (ID:90254).

Data Availability

The data used in this study is available at: <https://doi.org/10.5281/zenodo.15640474>.

References

- [1] E. G. Nawy, Fundamentals of high-performance concrete, John Wiley & Sons, 2000.
- [2] E. C. for Standardization, En 1992-1-1: Eurocode 2: Design of concrete structures - part 1-1: General rules and rules for buildings (2004).
- [3] K. L. Scrivener, K. M. Nematì, The percolation of pore space in the cement paste/aggregate interfacial zone of concrete, Cement and concrete research 26 (1) (1996) 35–40.
- [4] K. L. Scrivener, A. K. Crumbie, P. Laugesen, The interfacial transition zone (itz) between cement paste and aggregate in concrete, Interface science 12 (2004) 411–421.
- [5] Q. Chen, J. Zhang, Z. Wang, T. Zhao, Z. Wang, A review of the interfacial transition zones in concrete: Identification, physical characteristics, and mechanical properties, Engineering Fracture Mechanics 300 (2024) 109979.
- [6] Z. Hashin, P. Monteiro, An inverse method to determine the elastic properties of the interphase between the aggregate and the cement paste, Cement and Concrete Research 32 (8) (2002) 1291–1300.
- [7] J. Lee, G. L. Fenves, Plastic-damage model for cyclic loading of concrete structures, Journal of engineering mechanics 124 (8) (1998) 892–900.
- [8] J. Červenka, V. K. Papanikolaou, Three dimensional combined fracture–plastic material model for concrete, International journal of plasticity 24 (12) (2008) 2192–2220.
- [9] F. C. Caner, Z. P. Bažant, Microplane model m7 for plain concrete. i: Formulation, Journal of Engineering Mechanics 139 (12) (2013) 1714–1723.
- [10] I. Zreid, M. Kaliske, A gradient enhanced plasticity–damage microplane model for concrete, Computational Mechanics 62 (5) (2018) 1239–1257.

- [11] Z.-J. Yang, B.-B. Li, J.-Y. Wu, X-ray computed tomography images based phase-field modeling of mesoscopic failure in concrete, *Engineering Fracture Mechanics* 208 (2019) 151–170.
- [12] P. Wriggers, S. Moftah, Mesoscale models for concrete: Homogenisation and damage behaviour, *Finite elements in analysis and design* 42 (7) (2006) 623–636.
- [13] M. Nitka, J. Teichman, Meso-mechanical modelling of damage in concrete using discrete element method with porous itzs of defined width around aggregates, *Engineering Fracture Mechanics* 231 (2020) 107029.
- [14] E. A. Rodrigues, O. L. Manzoli, L. A. Bitencourt Jr, 3d concurrent multiscale model for crack propagation in concrete, *Computer Methods in Applied Mechanics and Engineering* 361 (2020) 112813.
- [15] G. Cusatis, D. Pelessone, A. Mencarelli, Lattice discrete particle model (ldpm) for failure behavior of concrete. i: Theory, *Cement and Concrete Composites* 33 (9) (2011) 881–890.
- [16] A. A. Munjiza, *The combined finite-discrete element method*, John Wiley & Sons, 2004.
- [17] M. Nitka, J. Teichman, A three-dimensional meso-scale approach to concrete fracture based on combined dem with x-ray μ ct images, *Cement and Concrete Research* 107 (2018) 11–29.
- [18] Y. Zhang, Q. Chen, Z. Wang, J. Zhang, Z. Wang, Z. Li, 3d mesoscale fracture analysis of concrete under complex loading, *Engineering Fracture Mechanics* 220 (2019) 106646.
- [19] R. Zhou, Y. Lu, A mesoscale interface approach to modelling fractures in concrete for material investigation, *Construction and Building Materials* 165 (2018) 608–620.
- [20] J. Zhang, H. Yang, Y. Zhang, Z. Wang, Z. Wang, X. Shu, Numerical investigations on the effect of reinforcement on penetration resistance of concrete slabs using a 3d meso-scale method, *Construction and Building Materials* 188 (2018) 793–808.
- [21] R. Zhou, Y. Lu, L.-G. Wang, H.-M. Chen, Mesoscale modelling of size effect on the evolution of fracture process zone in concrete, *Engineering Fracture Mechanics* 245 (2021) 107559.
- [22] P. C. Paris, G. C. Sih, Stress analysis of cracks, *ASTM stp 381* (1965) 30–81.
- [23] T. L. Anderson, T. L. Anderson, *Fracture mechanics: fundamentals and applications*, CRC press, 2005.
- [24] A. Hillerborg, M. Mod  er, P.-E. Petersson, Analysis of crack formation and crack growth in concrete by means of fracture mechanics and finite elements, *Cement and concrete research* 6 (6) (1976) 773–781.
- [25] M. Jirasek, Nonlocal models for damage and fracture: comparison of approaches, *International Journal of Solids and Structures* 35 (31-32) (1998) 4133–4145.
- [26] J. Xiao, W. Li, D. J. Corr, S. P. Shah, Effects of interfacial transition zones on the stress–strain behavior of modeled recycled aggregate concrete, *Cement and Concrete Research* 52 (2013) 82–99.
- [27] J. Xiao, W. Li, Z. Sun, D. A. Lange, S. P. Shah, Properties of interfacial transition zones in recycled aggregate concrete tested by nanoindentation, *Cement and Concrete Composites* 37 (2013) 276–292.
- [28] R. V. Silva, J. De Brito, R. K. Dhir, Establishing a relationship between modulus of elasticity and compressive strength of recycled aggregate concrete, *Journal of cleaner production* 112 (2016) 2171–2186.

- [29] M. Adamson, A. Razmjoo, A. Poursaei, Durability of concrete incorporating crushed brick as coarse aggregate, *Construction and building materials* 94 (2015) 426–432.
- [30] S. Narayanan, M. Sirajuddin, Properties of brick masonry for fe modeling, *American Journal of Engineering Research* 1 (2013) (2013) 6–11.
- [31] N. Benkemoun, M. Hautefeuille, J.-B. Colliat, A. Ibrahimbegovic, Failure of heterogeneous materials: 3d meso-scale fe models with embedded discontinuities, *International journal for numerical methods in engineering* 82 (13) (2010) 1671–1688.
- [32] S. A. Galindo-Torres, D. Pedroso, D. Williams, L. Li, Breaking processes in three-dimensional bonded granular materials with general shapes, *Computer Physics Communications* 183 (2) (2012) 266–277.
- [33] R. Zhou, Z. Song, Y. Lu, 3d mesoscale finite element modelling of concrete, *Computers & Structures* 192 (2017) 96–113.
- [34] F. Aurenhammer, Power diagrams: properties, algorithms and applications, *SIAM journal on computing* 16 (1) (1987) 78–96.
- [35] R. Szczelina, K. Murzyn, Dmg- α a computational geometry library for multimolecular systems, *Journal of Chemical Information and Modeling* 54 (11) (2014) 3112–3123.
- [36] W. contributors, Hyperplane separation theorem — wikipedia, the free encyclopedia, accessed: 2025-01-28 (2025).
URL https://en.wikipedia.org/wiki/Hyperplane_separation_theorem
- [37] A. Kaszynski, pyansys: Pythonic interface to MAPDL (Nov. 2021). doi:10.5281/zenodo.4009466.
URL <https://doi.org/10.5281/zenodo.4009466>
- [38] C. Geuzaine, J.-F. Remacle, Gmsh: A 3-d finite element mesh generator with built-in pre-and post-processing facilities, *International journal for numerical methods in engineering* 79 (11) (2009) 1309–1331.
- [39] J. Mašek, J. Květoň, J. Eliáš, Adaptive discretization refinement for discrete models of coupled mechanics and mass transport in concrete, *Construction and Building Materials* 395 (2023) 132243.
- [40] A. B. Yoo, M. A. Jette, M. Grondona, Slurm: Simple linux utility for resource management, in: *Job Scheduling Strategies for Parallel Processing*, Springer, 2003, pp. 44–60. doi:10.1007/10968987_3.
- [41] H. Tada, P. C. Paris, G. R. Irwin, The stress analysis of cracks, *Handbook*, Del Research Corporation 34 (1973) (1973).
- [42] S. Blasón, T. Werner, J. Kruse, M. Madia, P. Miarka, S. Seitzl, M. Benedetti, Determination of fatigue crack growth in the near-threshold regime using small-scale specimens, *Theoretical and applied fracture mechanics* 118 (2022) 103224.
- [43] R. Hurley, D. Pagan, An in-situ study of stress evolution and fracture growth during compression of concrete, *International Journal of Solids and Structures* 168 (2019) 26–40.
- [44] C. Perry, J. Gillott, The influence of mortar-aggregate bond strength on the behaviour of concrete in uniaxial compression, *Cement and Concrete Research* 7 (5) (1977) 553–564.

- [45] M. B. Cil, K. A. Alshibli, 3d evolution of sand fracture under 1d compression, *Géotechnique* 64 (5) (2014) 351–364.
- [46] M. B. Cil, K. A. Alshibli, P. Kenesei, 3d experimental measurement of lattice strain and fracture behavior of sand particles using synchrotron x-ray diffraction and tomography, *Journal of Geotechnical and Geoenvironmental Engineering* 143 (9) (2017) 04017048.
- [47] J. D. Eshelby, The determination of the elastic field of an ellipsoidal inclusion, and related problems, *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* 241 (1226) (1957) 376–396. doi:10.1098/rspa.1957.0133.
- [48] G. J. Rodin, Eshelby's inclusion problem for polygons and polyhedra, *Journal of the Mechanics and Physics of Solids* 44 (12) (1996) 1977–1995. doi:10.1016/S0022-5096(96)00066-X.
- [49] D. Wei, R. C. Hurley, L. H. Poh, D. Dias-da Costa, Y. Gan, The role of particle morphology on concrete fracture behaviour: A meso-scale modelling approach, *Cement and Concrete Research* 134 (2020) 106096.
- [50] J. D. Bass, et al., Elasticity of minerals, glasses, and melts, *Mineral physics and crystallography: A handbook of physical constants* 2 (1995) 45–63.
- [51] L. Guo, Z. Wu, L. Zhong, Y. Luo, Effect of aggregate characteristics on properties of cemented sand and gravel, *Science and Engineering of Composite Materials* 30 (1) (2023) 20220220.
- [52] H. Wadell, Sphericity and roundness of rock particles, *The Journal of Geology* 41 (3) (1933) 310–331.

Appendix A. Notes on mesh density

The accuracy and reliability of FEM-based numerical models are often influenced by mesh sensitivity, necessitating an appropriate mesh size to achieve both high-quality results and feasible computational efficiency. In this study, the level of physical discretization within each model is determined by the size and placement of the aggregates. The generated mesh consists of 3D tetrahedral elements, chosen for their compatibility with the complex geometries of the aggregates while ensuring numerical stability. Since the analysis is purely elastic, the primary objective is to accurately capture stress distributions without accounting for nonlinearities such as cracking or plasticity. The mesh is generated using Gmsh, which provides control over element quality and ensures a well-conditioned discretization. Additionally, mesh optimization (Mesh.Optimize) is applied to improve element shapes and enhance numerical stability. Mesh density is manually controlled through the maximum element size, $l_{\max}$, to impose sufficient refinement in the notch vicinity.

A mesh convergence study was performed to determine an optimal element size, ensuring that computed stress fields and displacement responses remained consistent across successive refinements. The final mesh configuration balances accuracy and computational cost, capturing essential features of the elastic stress distribution without excessive computational overhead. The optimal mesh size was verified before proceeding with further numerical analysis and was subsequently adopted. This mesh study was conducted using a three-point bending test model (see Section 4.2). Several progressively finer meshes were examined, as summarized in Table A.2.

Due to the inherent complexity of the model, the prescribed maximum element size cannot always be strictly maintained. Regions with intricate geometries or sharp features require finer elements to ensure proper mesh generation. In such cases, the meshing algorithm automatically reduces element size to maintain quality and prevent element distortion. As a result, while the specified maximum element size serves as a guideline, local refinements occur where necessary to accommodate the model's geometric constraints.

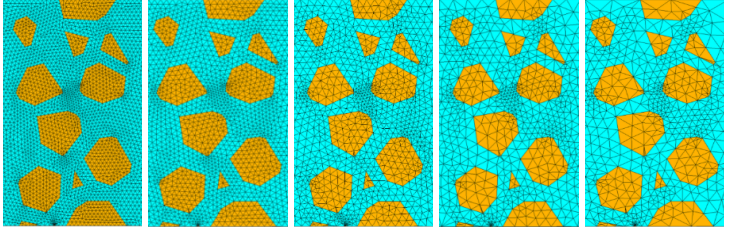
$l_{\max}$	n.o. Elements [-]	n.o. DOF [-]	1 mm	2 mm	3 mm	5 mm	10 mm
1 mm	$2\,150 \times 10^3$	$1\,140 \times 10^3$					
2 mm	435×10^3	305×10^3					
3 mm	232×10^3	696×10^3					
5 mm	160×10^3	480×10^3					
10 mm	158×10^3	474×10^3					

Table A.2: Mesh study: midspan mesh of the TPB Model.

The study was conducted on a three-point bending beam model, and the obtained stress distribution was compared to the analytical solution. Convergence was examined by progressively refining the mesh from coarser to finer resolutions. First row of Figure A.24 shows the respective normal stress histograms of each studied mesh. The second row shows the absolute error, $\epsilon(\sigma_x)$, of each model envelopes respective to the analytical solution.

The results indicate that a 1 mm element size was excessively fine, leading to a substantial increase in computational cost without a proportional improvement in accuracy. Conversely, the 2 mm mesh provided sufficient precision in capturing stress distributions while maintaining a reasonable computational expense. The differences were within acceptable limits, confirming that the chosen discretization effectively captured the expected stress behavior. Following the convergence study, a mesh with a maximum element size of 2 mm was selected for subsequent analyses. The 2 mm mesh provided sufficient precision in capturing stress distributions while avoiding excessive computational costs associated with finer discretizations. A further reduction in element size would have significantly increased the computational burden without yielding substantial improvements in accuracy.

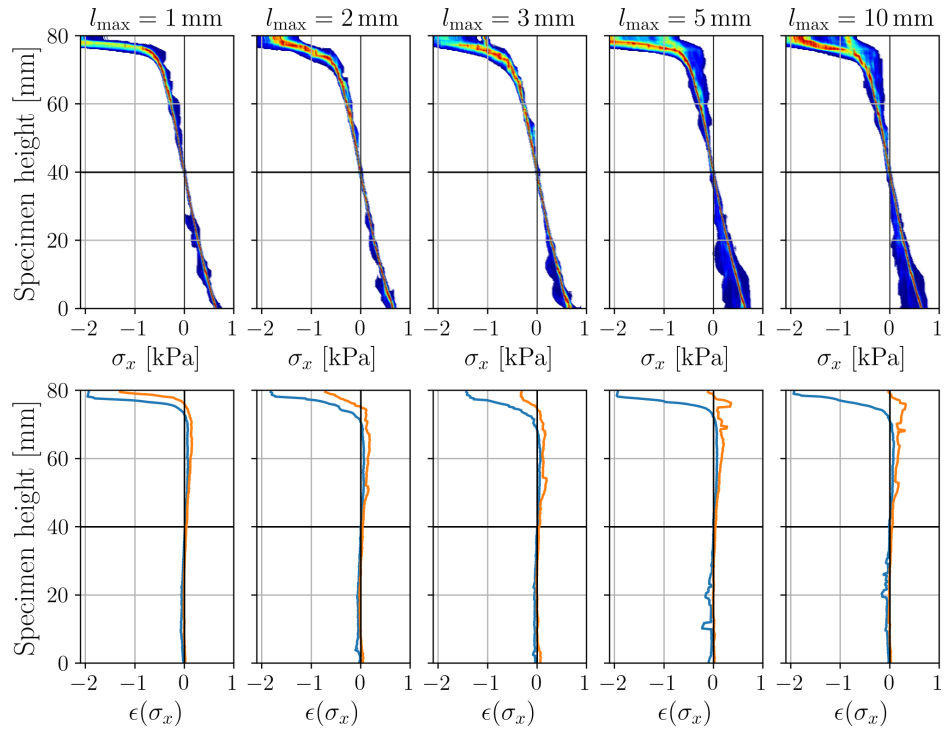


Figure A.24: Mesh error study

Appendix B. ANSYS to GMSH Mesh Pipeline

```
from ansys.mapdl.core import launch_mapdl
from ansys.mapdl.reader import save_as_archive
import pyvista as pv
import os
import gmsh

# Launch MAPDL instance
mapdl = launch_mapdl(
    jobname="pyzx",
    loglevel="ERROR",
    print_com=True,
    log_apdl=True,
    override=True,
    nproc=4
)

# Load ANSYS macro containing volume geometry
mapdl.input("path_to_ansys_volume_model_macro")

# Export geometry to IGES
mapdl.cdwrite("SOLID", "modelout", "iges")

# Initialize GMSH
gmsh.initialize()
gmsh.option.setNumber("General.NumThreads", 4) # Use 4 threads
gmsh.option.setNumber("General.Terminal", 1) # Enable terminal output

# Import IGES geometry
v = gmsh.model.occ.importShapes("modelout.iges") # Returns tags of imported shapes
gmsh.model.occ.synchronize() # Synchronize GMSH CAD kernel

# Generate and optimize 3D mesh
gmsh.model.mesh.generate(3)
gmsh.model.mesh.optimize()
gmsh.write("from_gmsh.vtk") # Export mesh for PyVista

# Extract topology and adjacency info
topology = []
adj = []
for e in gmsh.model.get_entities(3):
    dim, tag = e
    up, down = gmsh.model.getAdjacencies(dim, tag) # 'down' = connected nodes
    elemTypes, elemTags, elemNodeTags = gmsh.model.mesh.getElements(dim, tag)
    adj.append(down)
    topology.append(elemTags)
```

```

gmsh.finalize() # Cleanly exit GMSH

# Load mesh in PyVista, scale and resave into ANSYS *.db format
mesh = pv.read("from_gmsh.vtk")
mesh.points /= 1000.0 # Scale coordinates

# Save mesh in MAPDL-readable archive and read into MAPDL
save_as_archive("tmp.db", mesh)
response = mapdl.cdread("db", "tmp.db")

# Merge duplicate nodes. Volumes have already been subtracted in Ansys.
# Therefore, subtracting them again in GMSH to obtain conforming mesh boundaries is unnecessary.
mapdl.allsel()
mapdl.nummrg("node", "all", 1e-7)

# Assign aggregate material and type
mapdl.allsel()
mapdl.emodif("all", "type", 11)
mapdl.emodif("all", "mat", 11)

# Identify largest volume for matrix
# Assign matrix material and type
mortarVolu = max(range(len(adj)), key=lambda i: len(adj[i]))
start_tag = topology[mortarVolu][0][0]
end_tag = topology[mortarVolu][0][-1]
mapdl.run(f"FLST,5,{start_tag},2,ORDE,2")
mapdl.run(f"FITEM,5,{start_tag}")
mapdl.run(f"FITEM,5,-{end_tag}")
mapdl.run("ESEL,S,,P51X")

# Assign matrix material and type
mapdl.emodif("all", "type", 22)
mapdl.emodif("all", "mat", 22)

# Save MAPDL model
mapdl.allsel("all")
mapdl.run("SAVE,modelname','','dirname'") # Replace with actual path

```

Listing 2: ANSYS to GMSH Mesh Pipeline